estar for quantifying ecological stability

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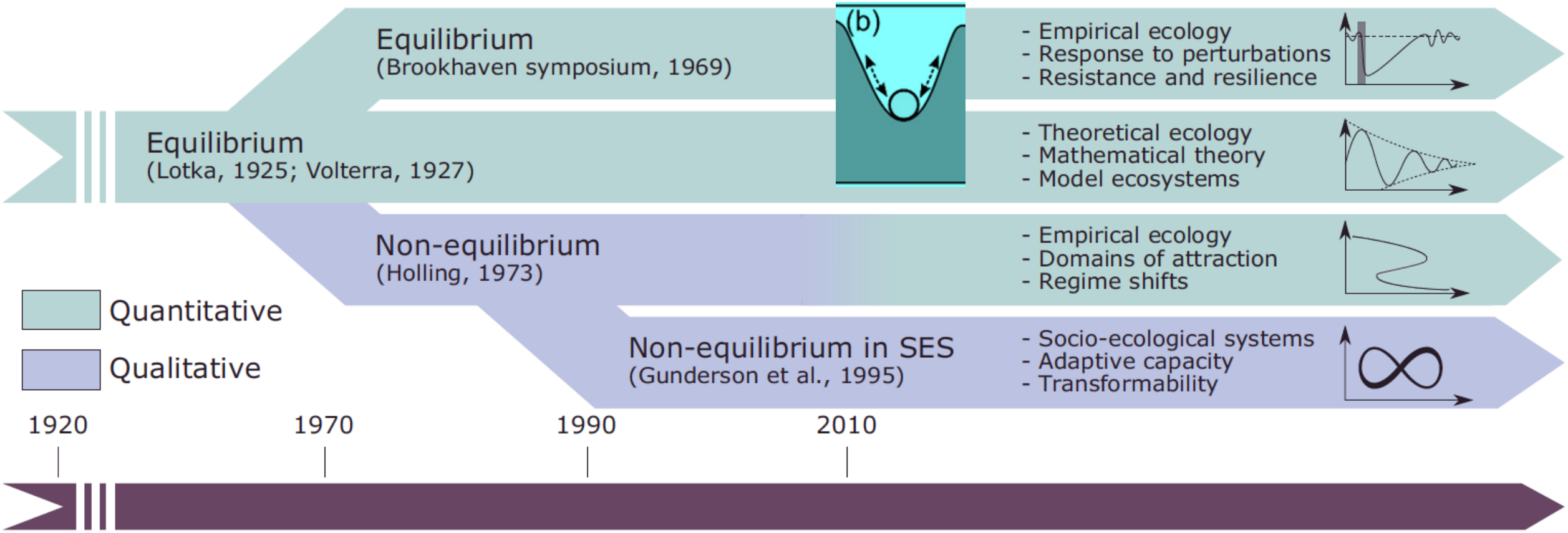
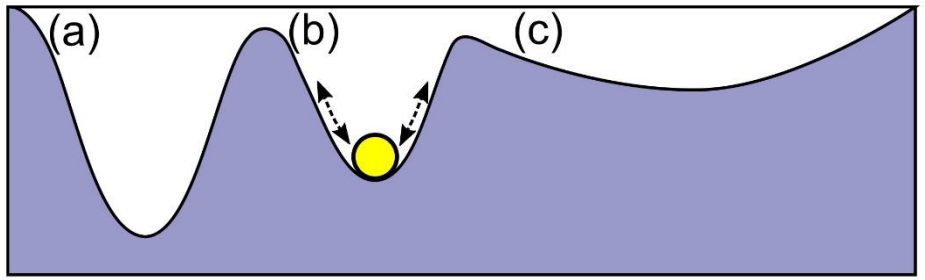
Stability: a key concept in ecology and evolution





Van Meerbeek et al (2021) J Ecol

Stability: a controversial concept

Oecologia (1997) 109:323–334 © Springer-Verlag 1997

Volker Grimm · Christian Wissel

Babel, or the ecological stability discussions: an inventory and analysis of terminology and a guide for avoiding confusion

- Found 163 definitions of 70 different stability concepts in literature

Grimm & Wissel (1997)

Stability: a controversial concept

Oecologia (1997) 109:323–334

Volker Grimm · Christian Wissel

Babel, or the ecological stability discussi
of terminology and a guide for avoiding co

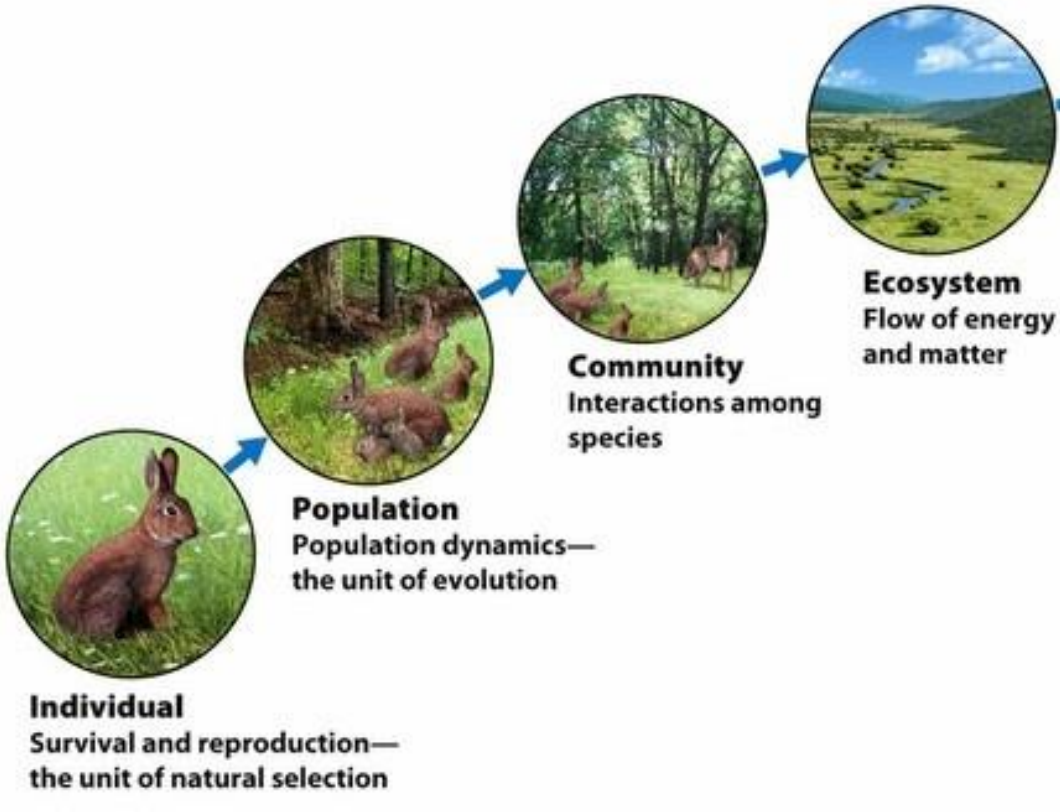
- Found 163 definitions of 70 different stability concepts in literature

Stability term and definition	Authors who use the term in the first column in more or less the same way	Terms with definition column
(1) **Constancy:** Staying essentially unchanged	Connell and Sousa 83:97 Gigon 83:97 Harrison 79:661 Lewontin 69:21 Orians 75:141 Remmert 89:286	Biomass stability – King and Pimm 1983:329 Ecological stability* – Zwölfer 78:15 Functional stability – Rejmánek 92:455 Perceived stability – Begon et al. 90:802 Persistence – Rahel 90:328 Stability* – Haber 79:24 Stability – Murdoch 70:497 Stability – Putman and Wratten 85:338 Temporal stability – Preston 69:9
(2) **Resilience:** Returning to the reference state (or dynamic) after a temporary disturbance	Harrison 79:660 Leps et al. 82:54 Putman and Wratten 85:339 Ulrich 92:181 Westman 78:705	Stability – Hallet 91:383 Stability – Holling 73:17 Stability – Pimm 84:322 Stability – Steele 74:180 Adjustment – Connell and Sousa 83:790 Connective stability – Siljak 74:280 Elasticity – Gigon 83:98 Elasticity* – Remmert 84:286 [Global, local] stability – Begon et al. 90:792 Mathematical stability – Danielson and Stenseth 92:83 Regulation – Murdoch 70:497 Resiliency – Kuss and Hall 91:715 Species deletion stability – Pimm 80:142
(3) **Persistence:** Persistence through time of an ecological system	Allen 83:4 Armstrong and McGhee 76:320 Botkin and Sobel 75:629 Connell and Sousa 83:791 DeAngelis and Waterhouse 87:7 Estberg and Patten 76:151 Harrison 79:660 Hastings 88:1666 Strong 90:421 Warner and Chesson 85:772	Stability – Begon et al. 90:792 Stability – Chesson and Huntly 89:293 Stability – Connell and Slatyer 77:1129 Stability – Crowley 92:246 Stability – Preston 69:7 Stability – Roff 74:246 Stability – Wu 76:156 Ecological stability – Nisbet and Gurney 82:10 Ecological stability – Wu 77:347 Essential stability – Wu 77:352

Grimm & Wissel (1997)

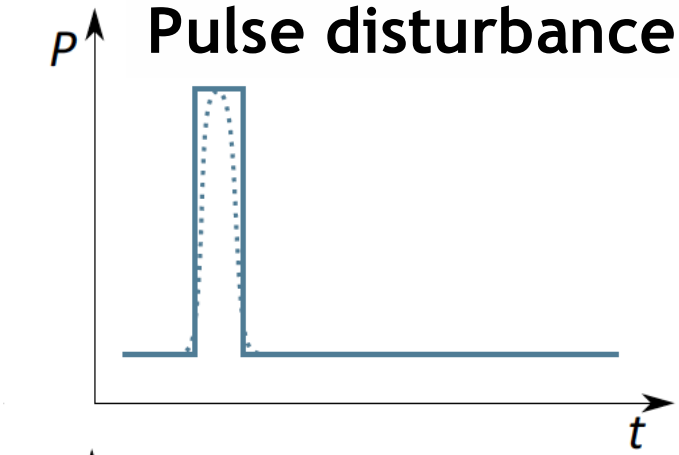
Some definitions

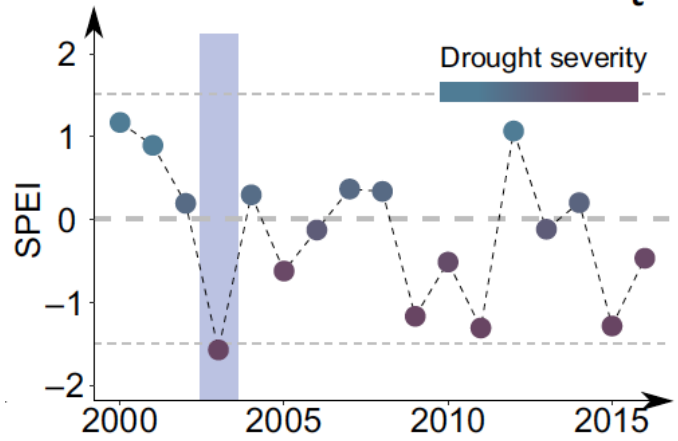
- System: here usually a population, community or ecosystem
- State variable is a variable that is used to quantify stability properties of a system. Examples of state variables are abundance (population level) and species richness or total abundance (community level)
- Baseline / reference dynamics – dynamics of the undisturbed system. Can be studied in a separate system or in the focal system prior to disturbance



Types of disturbances

- Disturbance is a change in the biotic or abiotic environment that alters the structure and dynamics of a system (Donohue et al. 2016).





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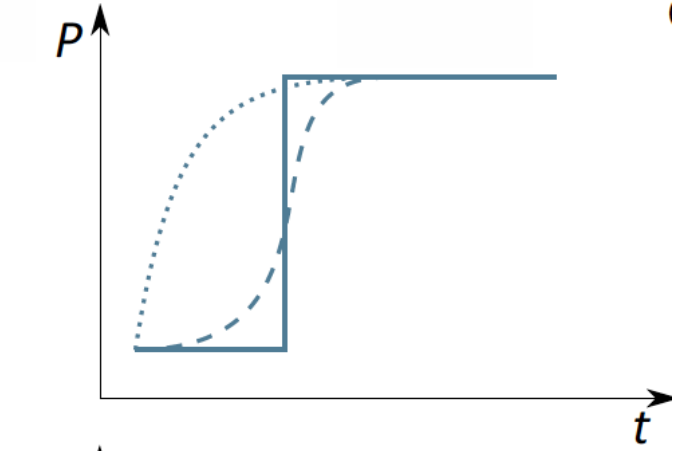
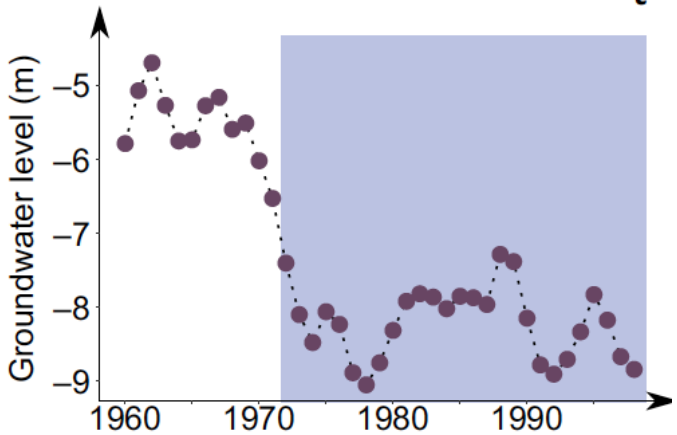
Types of disturbances

Press disturbance





Arise abruptly and then reach a constant level that is maintained

E.g. deforestation, habitat loss, decline in groundwater level

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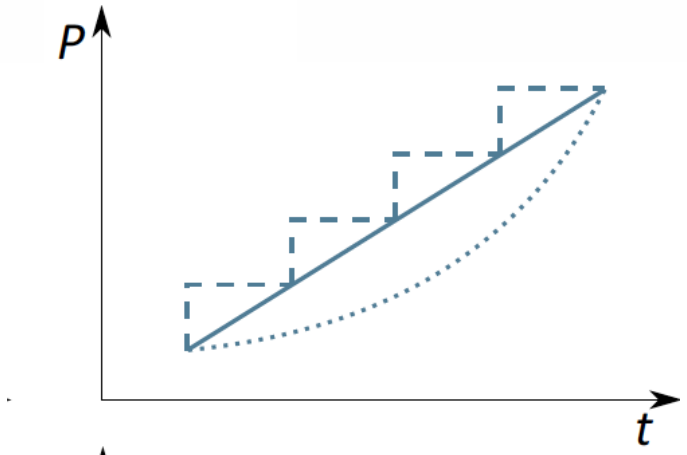
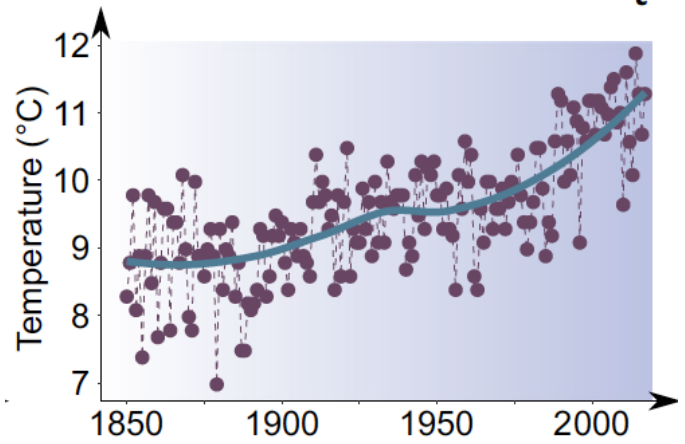
Types of disturbances

Characterised by a long-term and continuous change of the environment
E.g. climate change

Ramp disturbance





Van Meerbeek et al (2021) J Ecol

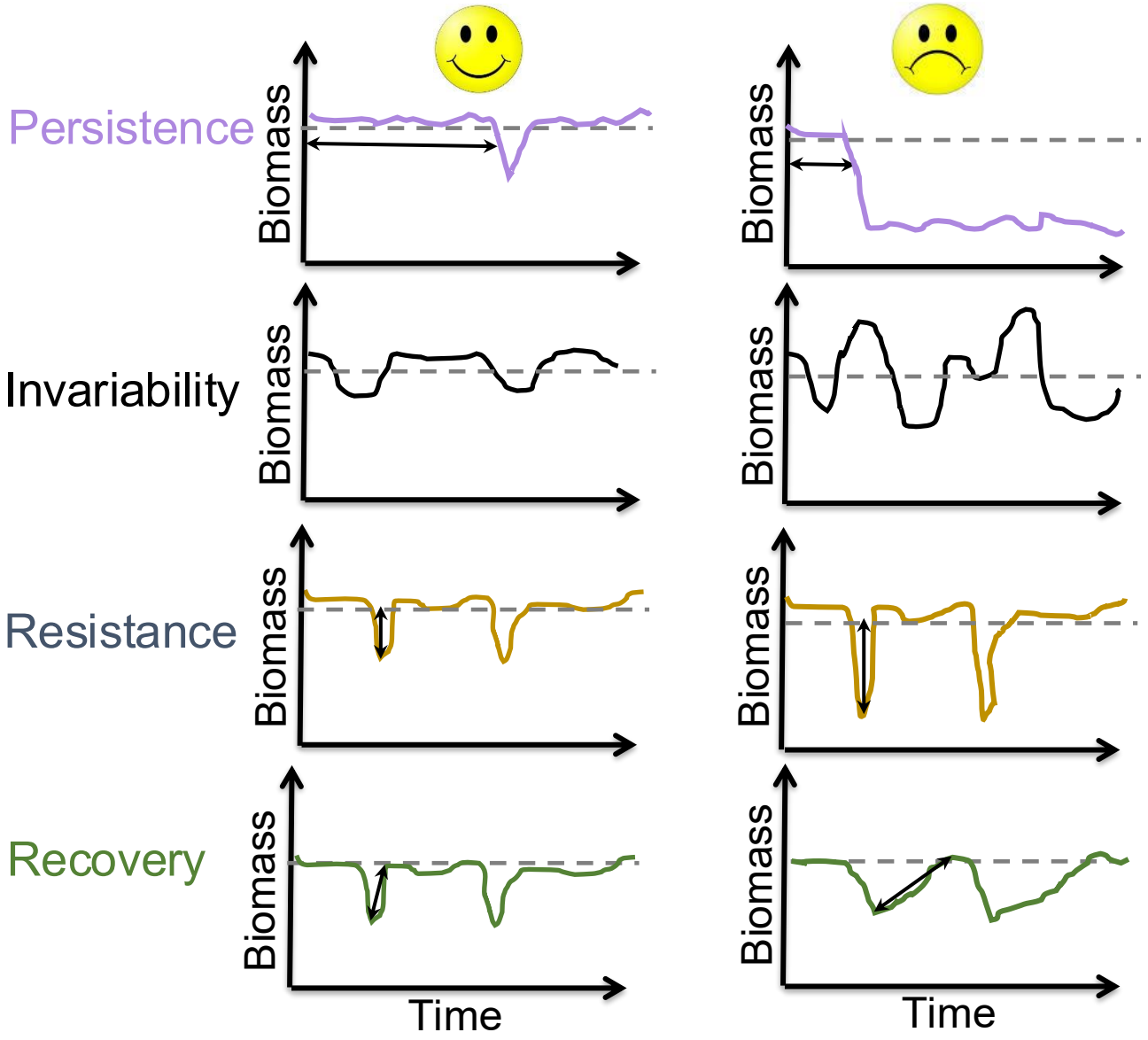
Reasons for controversy

Iain Donohue

- ‘stability is a multidimensional concept that tries to capture the different aspects of the dynamics of the system and its response to perturbations‘ (Donohue et al. 2016 EcolLets)
- Different definitions of stabiilty relate to different aspects of stability

Donohue et al. 2016 EcolLets

Multidimensional character of stability



- existence of a system through time as an identifiable unit
- The temporal constancy of a state variable following the disturbance, usually measured as the inverse of CV of a state variable
- Amount of change in a state variable following a disturbance
- Capacity of a system to return to its pre-disturbed state (different aspects: rate of recovery, recovery extent)

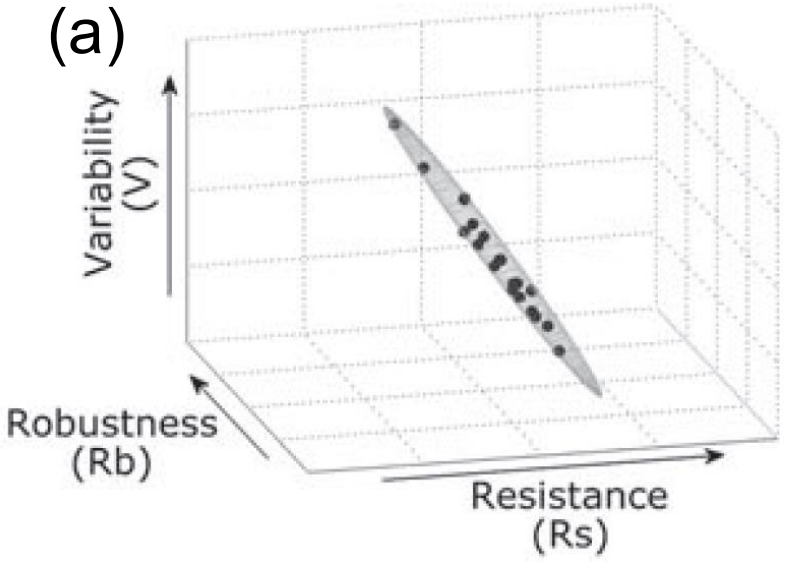
Oliver et al. 2015, Hodgson et al. 2015, Pimm 1984, Grimm & Wissel 1997

Different stability aspects do not necessarily correlate

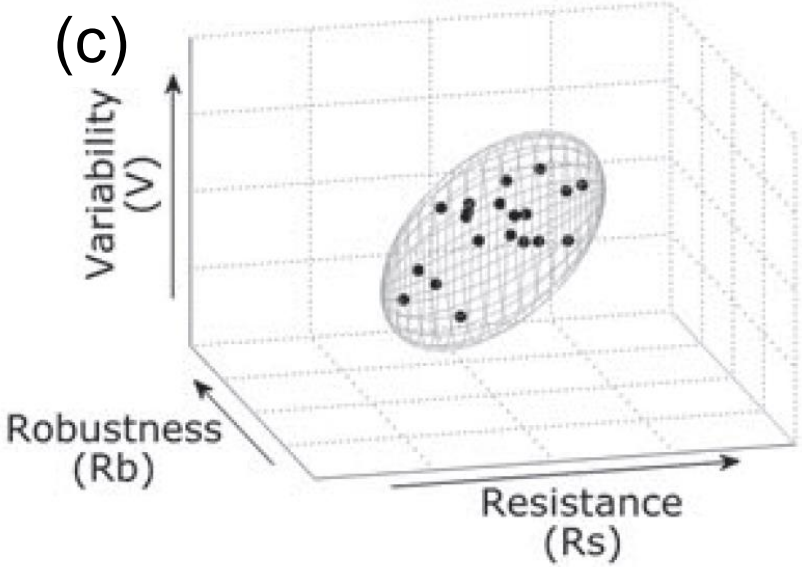
High correlation –
low dimensionality

Low correlation -
high dimensionality

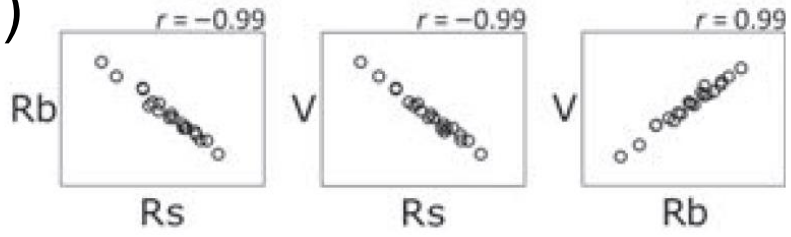
(a)



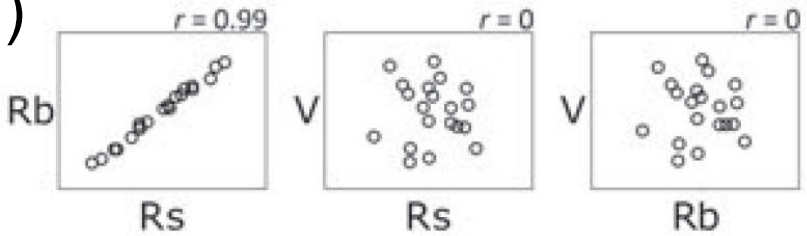
(c)



(b)

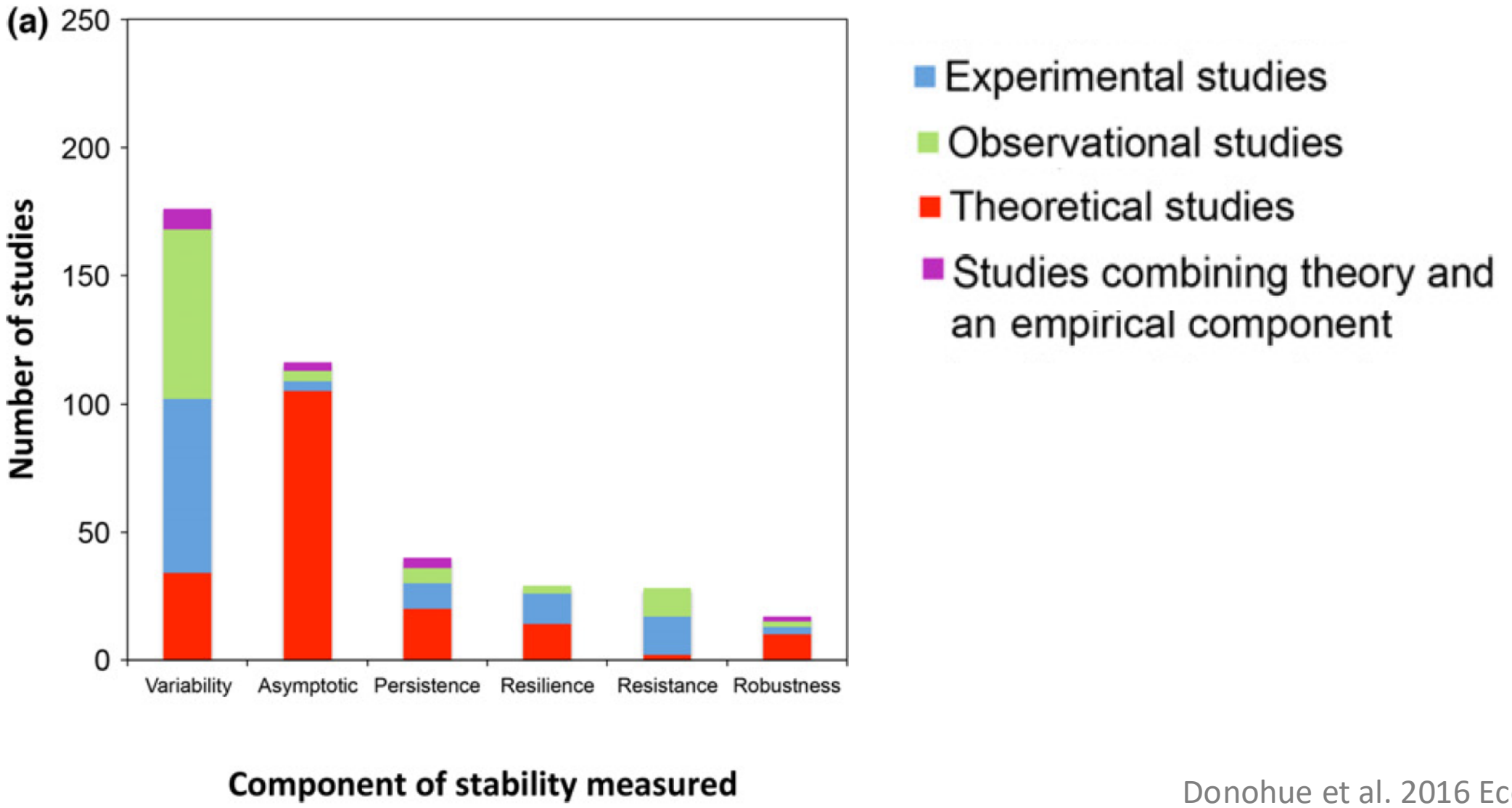


(d)



Donohue et al. 2013 EcolLets, Polazzo & Rico 2021 EcolLets, Radchuk et al. 2019 EcolLets

Empiricists and theoreticians measure different aspects of stability



Donohue et al. 2016 EcolLets

No single tool to quantify different stability metrics

Figueiredo et al. 2026 MEE

doi/10.1111/2041-210x.70265

Functional metrics

- Invariability
- Resistance
- Rate of recovery
- Extent of recovery
- Persistence
- Overall Ecological Vulnerability

Functional metrics: **type** argument

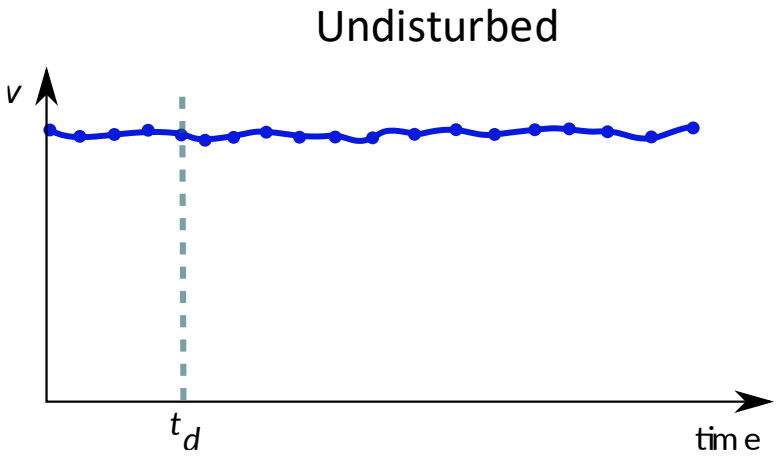
“functional”

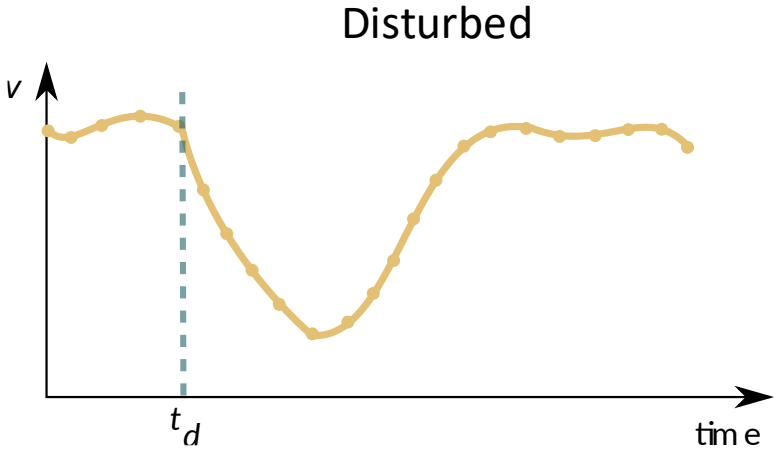
- Population abundance
- Trait value
- Ecosystem function

“compositional”

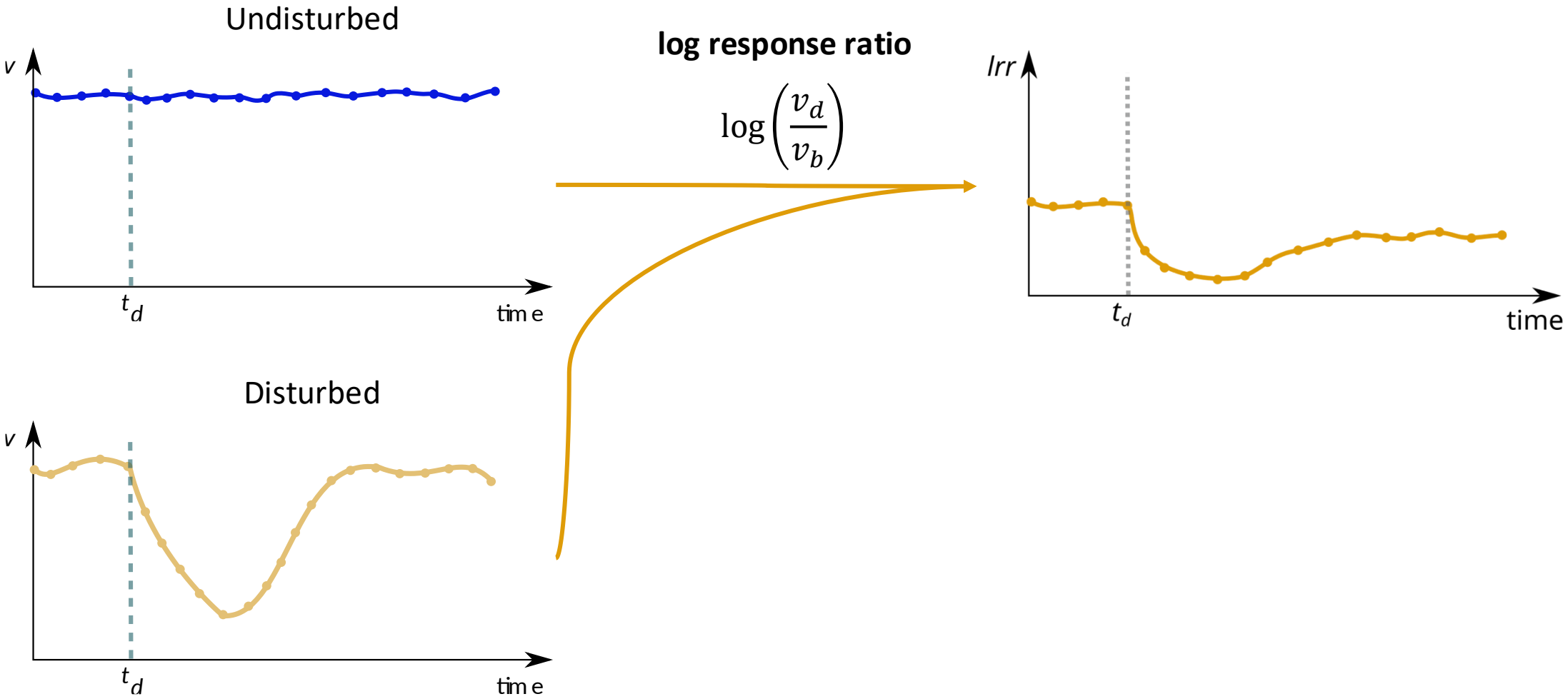
- Community dissimilarity

Functional metrics: response

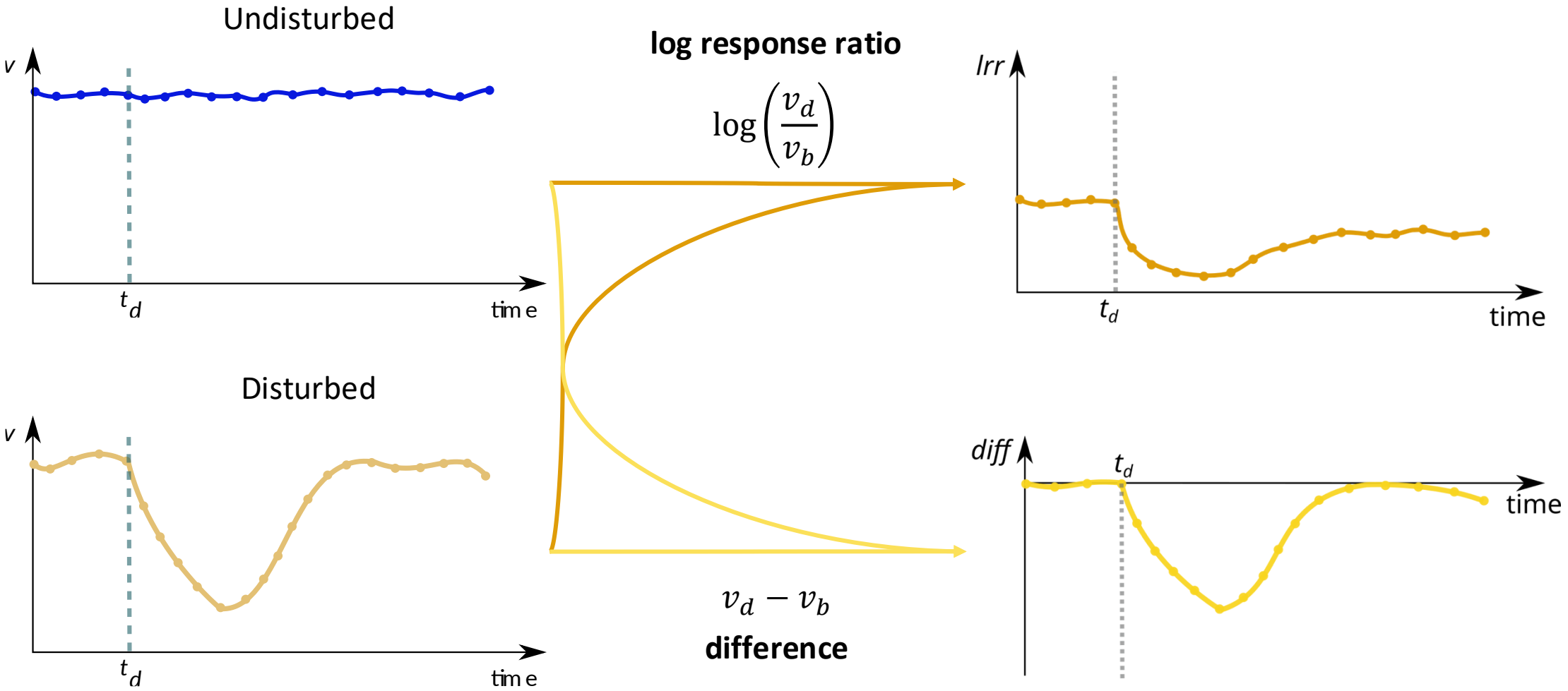




Functional metrics: response



Functional metrics: response



Data input, functional: **b**

time	baseline	disturbed
t*d*		

time	baseline
t*d*	

time	disturbed
t*d*	

Data input, functional: **b**

time	baseline	disturbed
td		

time	baseline
td	

time	disturbed
td	

b_tf

time	disturbed
td	

Data input, functional: **b**

time	baseline	disturbed
td		

time	baseline
td	

time	disturbed
td	

b_tf

time	disturbed
td	

b_data: name of baseline dataframe
vb_i: name of baseline state variable
tb_i: name of baseline time variable

d_data: name of disturbed dataframe
vd_i(vb_i): name of disturbed state variable
td_i: name of disturbed state variable
b_tf: time frame of baseline

Data input, compositional

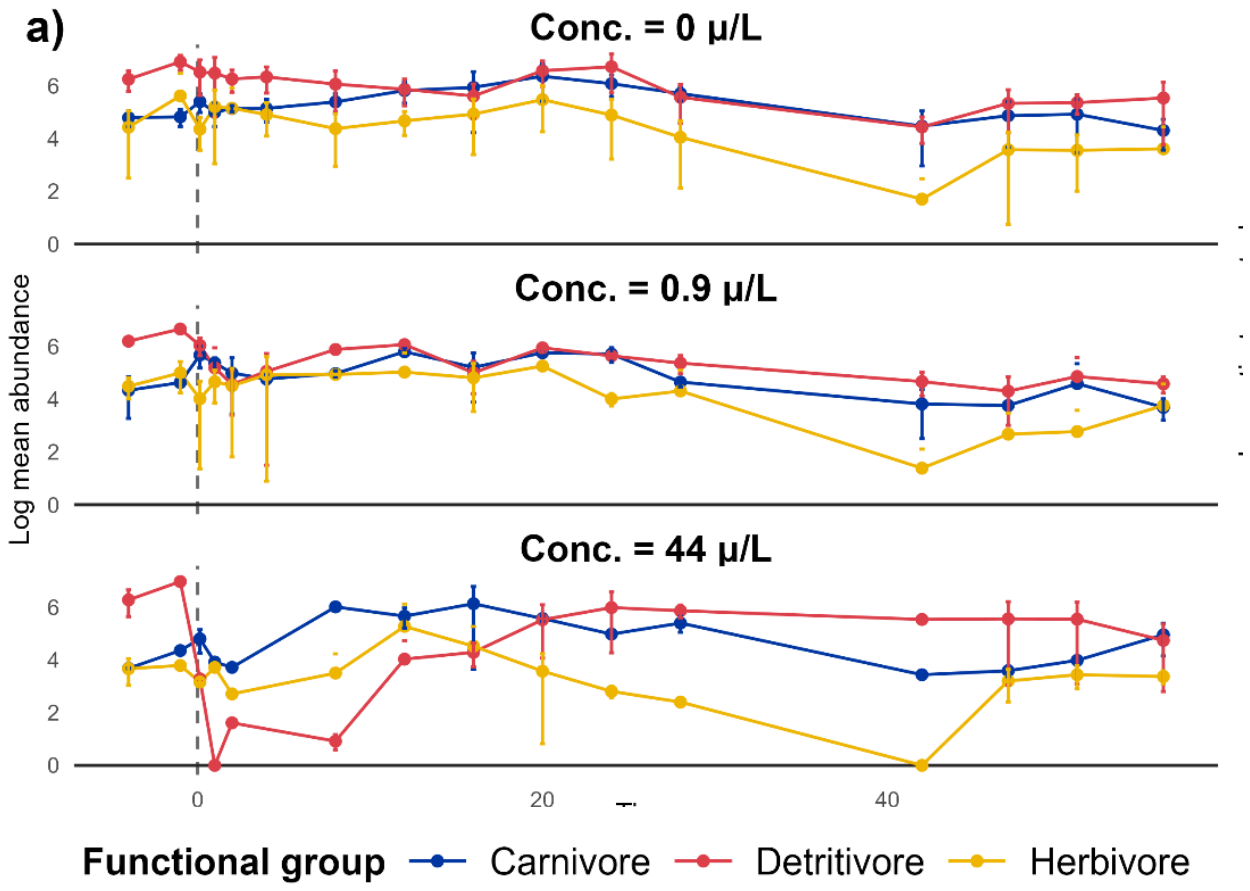
time	sp1	sp2	sp3	sp4

time	sp1	sp2	sp3	sp4

comm_b: name of the baseline community data
comm_d: name of the disturbed community data
comm_t: name of time variable
metric_tf
method: dissimilarity index
binary: presence/absence standardization

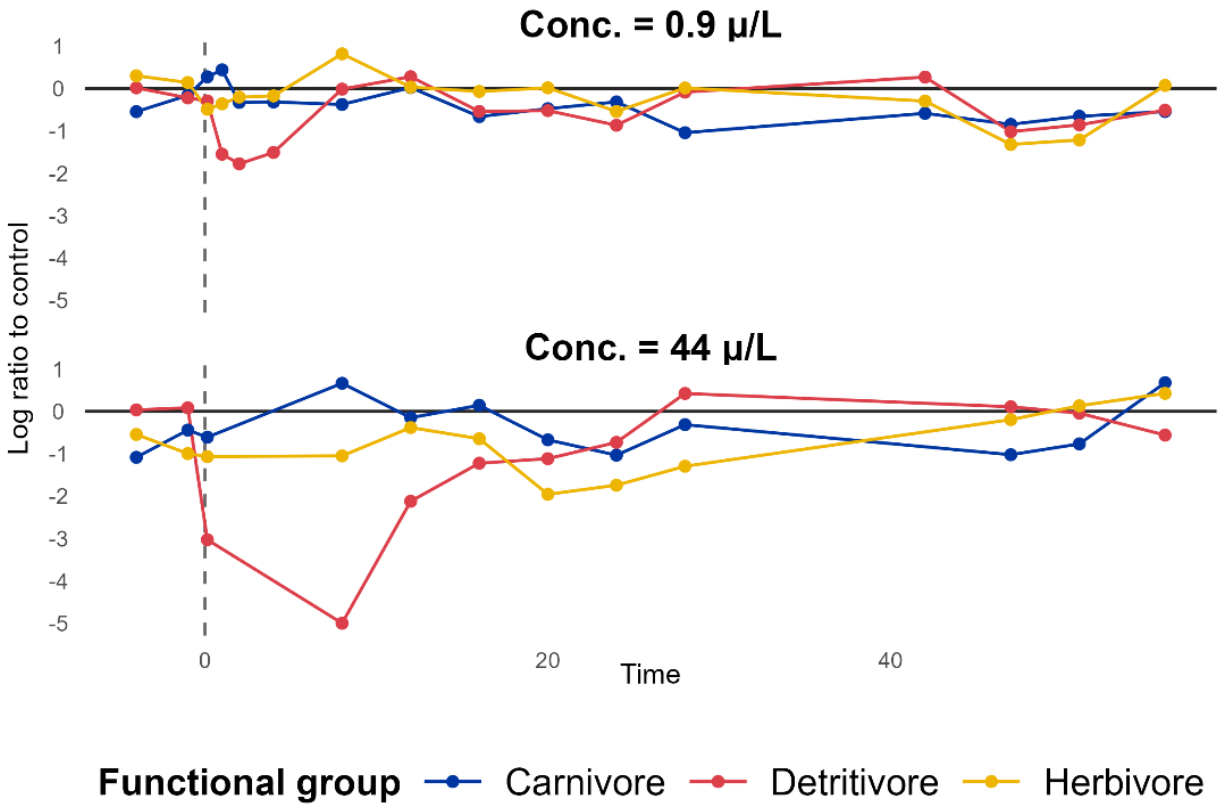
vegan

Data input

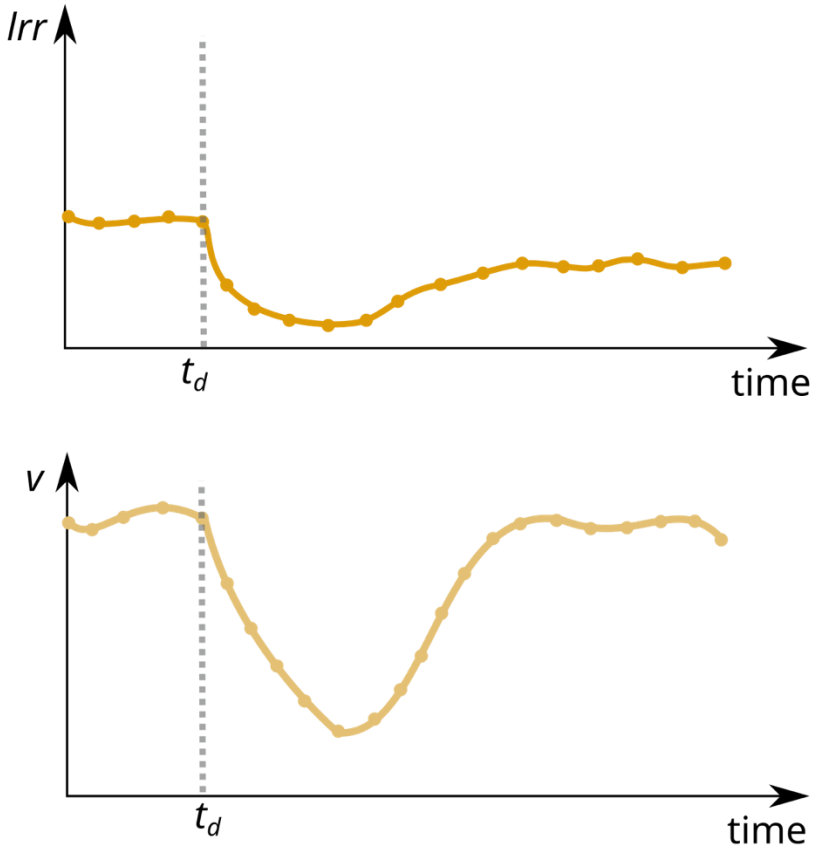


- Aquatic macroinvertebrates species into funct. groups
- 56 weeks observation
- Two concentrations of the chlorpyrifos insecticide

Data input


Conc. = 0.9 µ/L
Conc. = 44 µ/L
Log ratio to control
1
0
-1
-2
-3
-4
-5
0
20
40
Time
Functional group
Carnivore
Detritivore
Herbivore

invariability()




```
invariability(
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        mode = "lm_res",
        metric_tf = c(0.14, 56)),
        vd_i = "statvar_db",
        td_i = "time",
        d_data = aquacomm_resps,
        vb_i = "statvar_bl",
        tb_i = "time",
        b_data = aquacomm_resps)
```
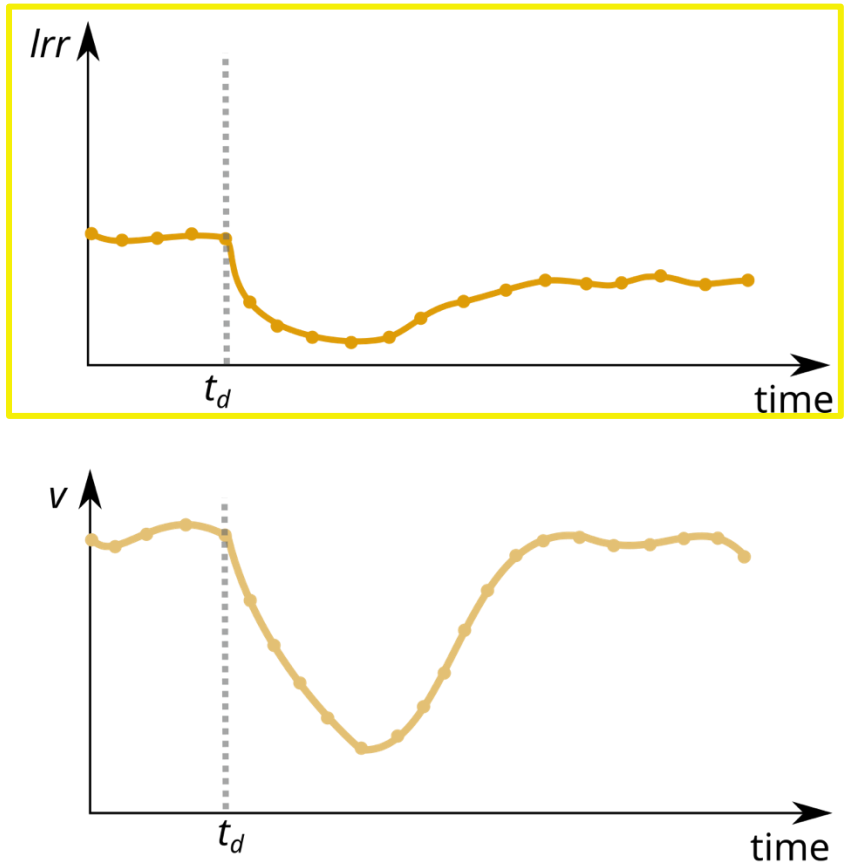

invariability()

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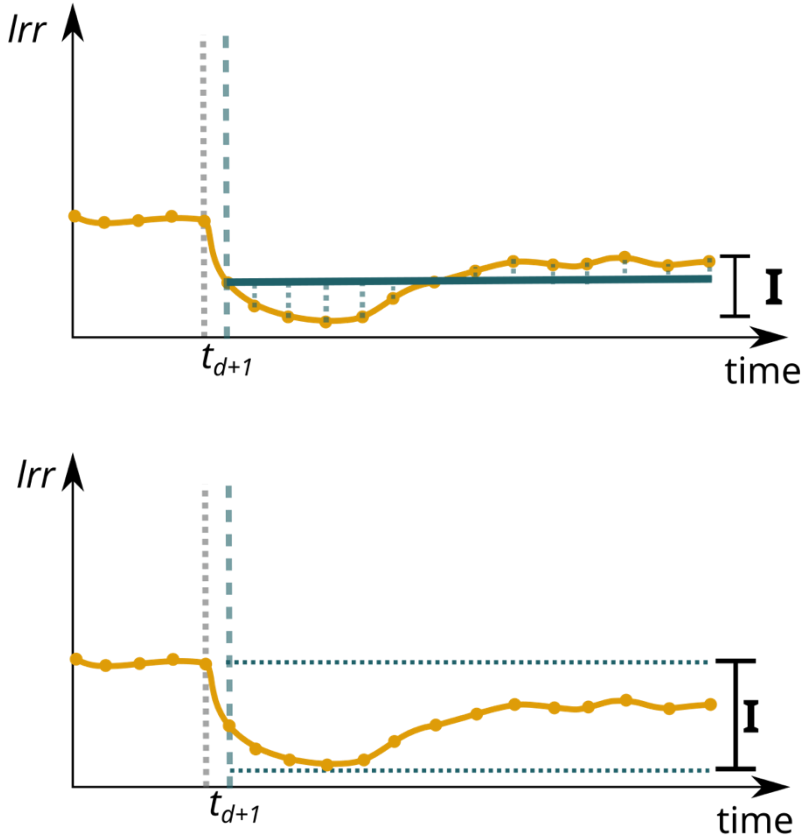
        vb_i = "statvar_bl",
        tb_i = "time",
        b_data = aquacomm_resps)
```

invariability()




```
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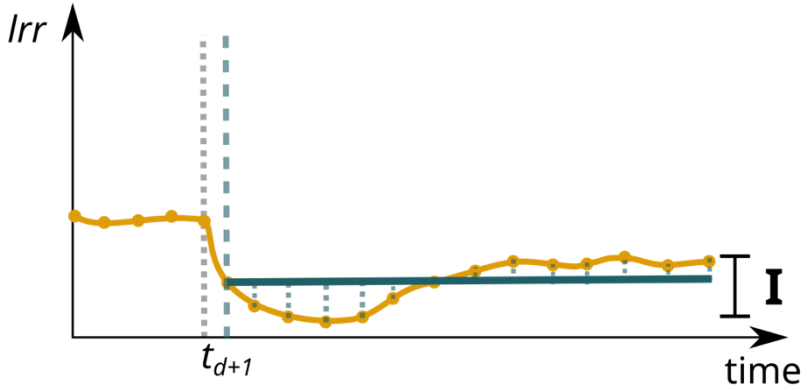
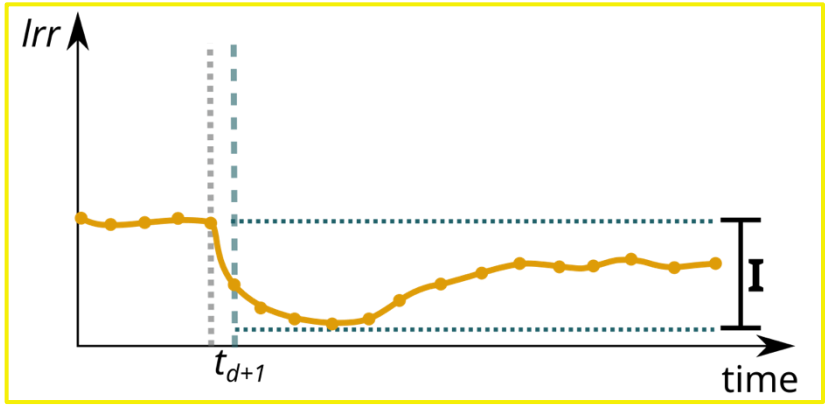
        vb_i = "statvar_bl",
        tb_i = "time",
        b_data = aquacomm_resps)
```

invariability()






```
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        mode = “cv",
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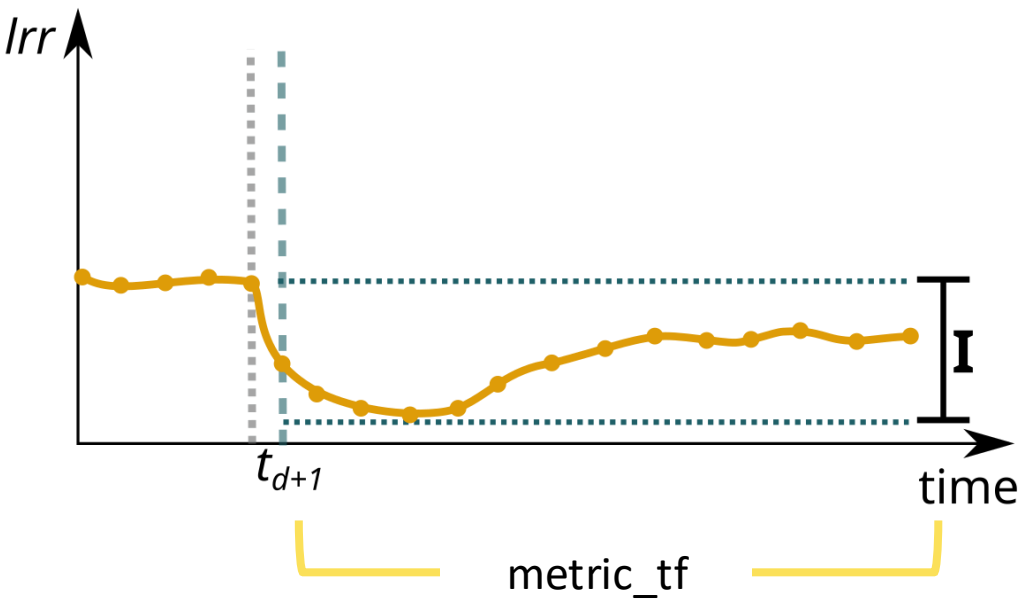
        td_i = "time",
        d_data = aquacomm_resps,
        vb_i = "statvar_bl",
        tb_i = "time",
        b_data = aquacomm_resps)
```

invariability()

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        tb_i = "time",
        b_data = aquacomm_resps)
```
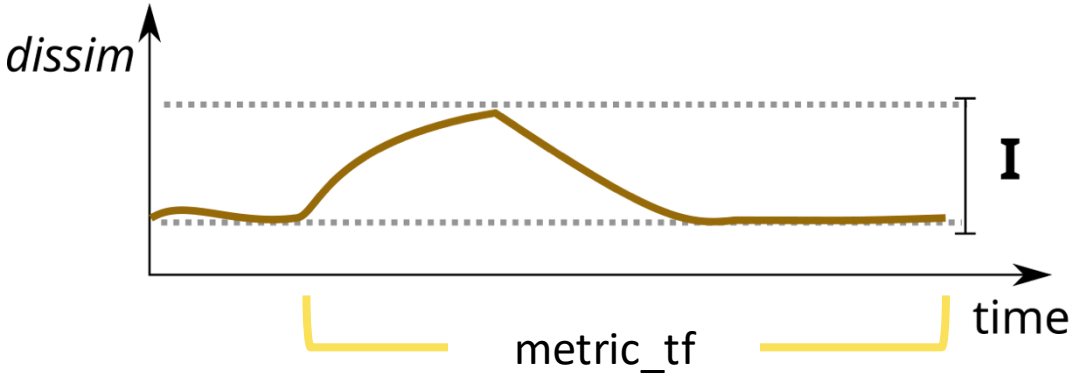
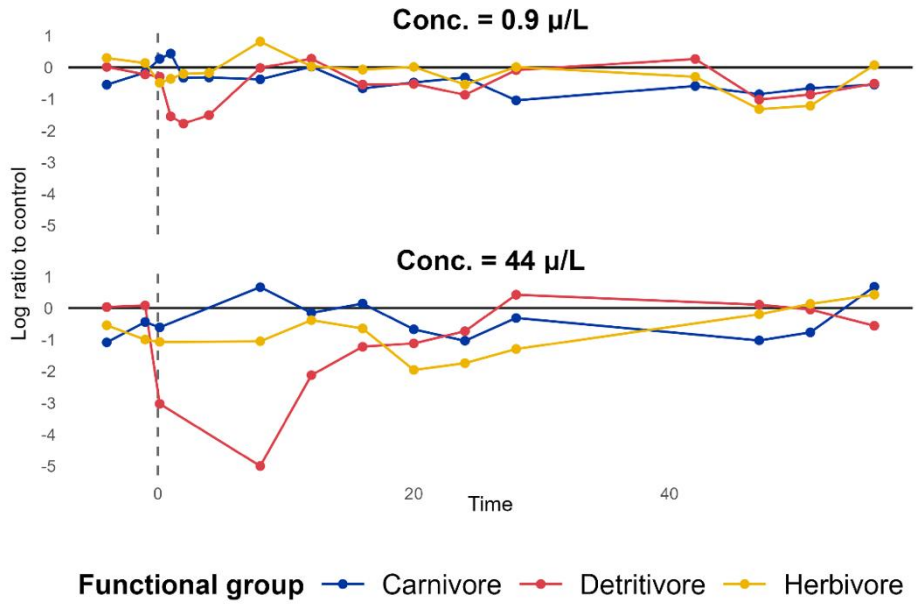

invariability()




```
invariability(
        type = "compositional",
        metric_tf = c(0.14, 56),
        method = "bray",
        binary = "FALSE“,
        …)
```

invariability()

Interpretation: The larger in magnitude, the higher the stability



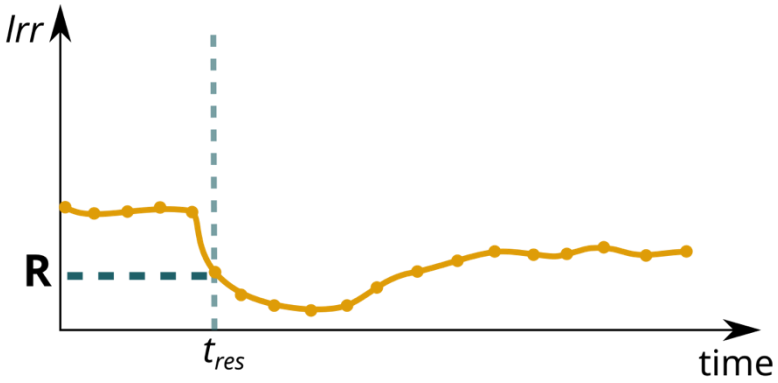

```
invariability(
  type = "functional",
  mode = “cv",
  response = "lrr",
  …)
```

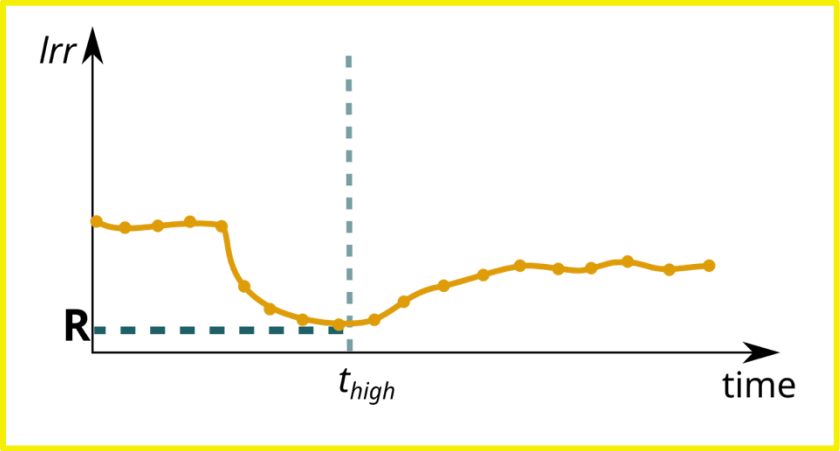
Group	Low	High
Carn	0.54	0.55
Detr	0.65	0.76
Herb	0.29	0.78

```
invariability(
  type = "compositional",
  …)
```

Low	High
6.46	5.87

resistance()



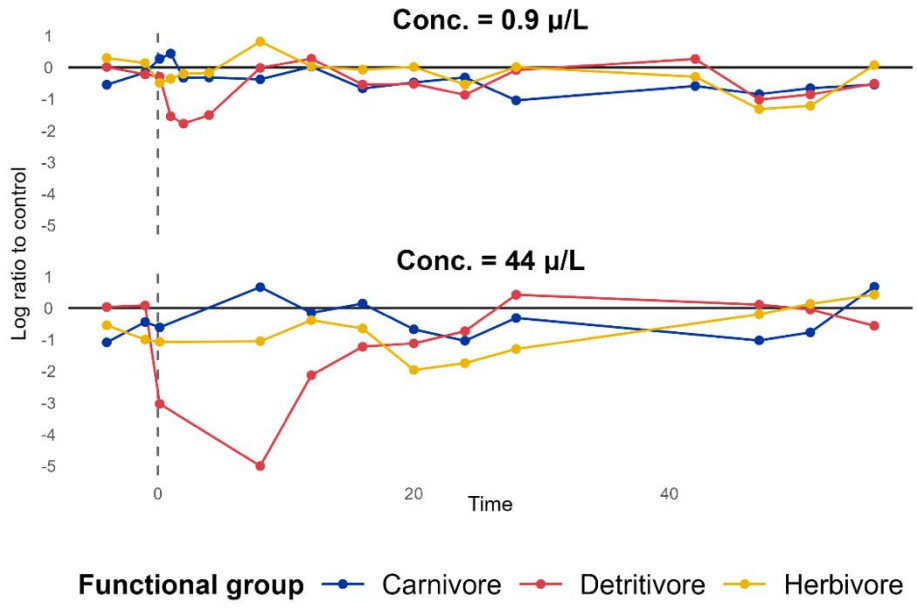



```
resistance(
        ...,
        res_time = "defined",
        res_t = 1,
        ...)
```

```
resistance(
        ...,
        res_time = "max",
        res_tf = c(0.14, 28),
        ...)
```

resistance()

Interpretation: Maximum resistance at 0. Smaller or higher values indicate under- or overcompensation



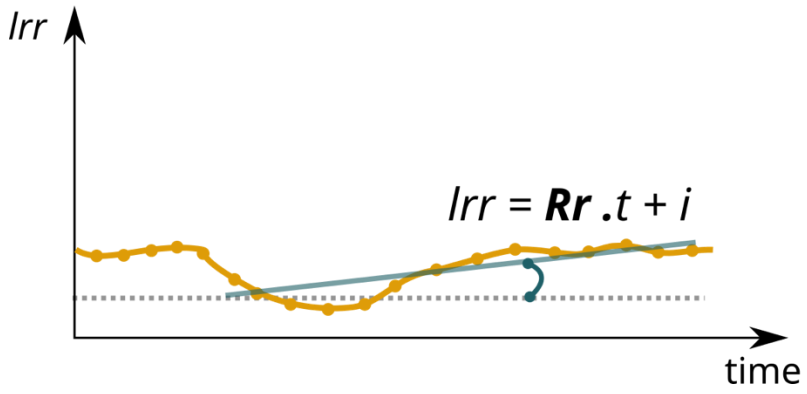

```
resistance(
  res_time = "max",
  res_tf = c(0.14, 28),
  …)
```

Group	Low	High
Carn	-1.01	-1.07
Detr	-2.09	-5.71
Herb	0.35	-2.99

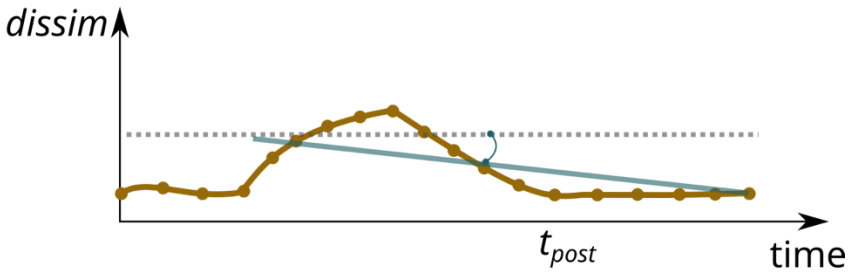
```
resistance(
  type = "compositional",
  …)
```

Low	High
0.64	0.84

recovery_rate()



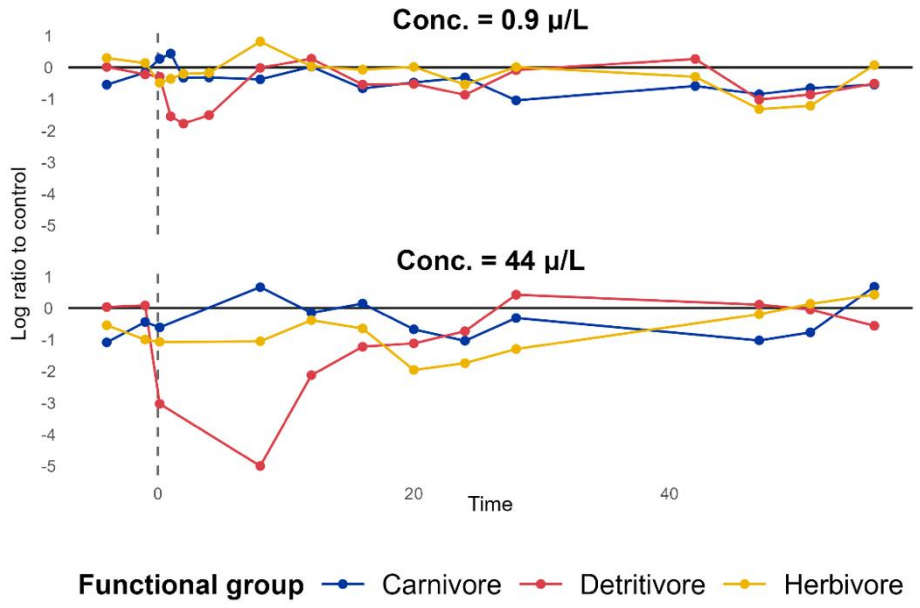

```
recovery_rate(
        type = "functional",
        response = "lrr",
        b = "input",
        metric_tf = c(28, 56),
        ...)
```




```
recovery_rate(
        type = "compositional",
        metric_tf = c(28, 56),
        ...)
```

recovery_rate()

Interpretation: The higher |Rr|, the faster the response.



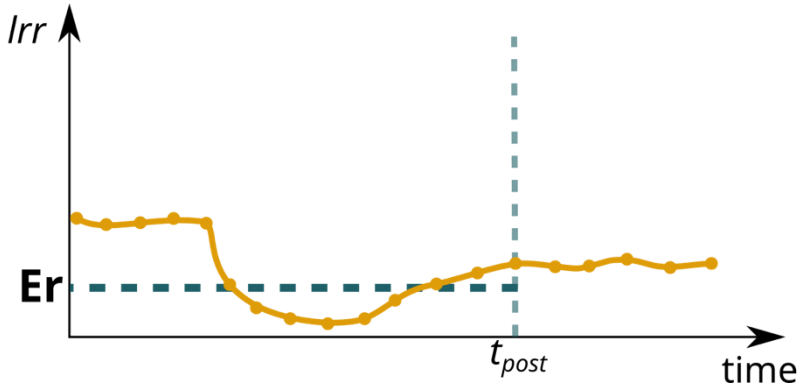

```
recovery_rate(
  type = "functional",
  response = "lrr",
  b = "input",
  metric_tf = c(28, 56),
  ...)
```

Group	Low	High
Carn	0.02	0.02
Detr	-0.03	-0.03
Herb	-0.02	-0.06

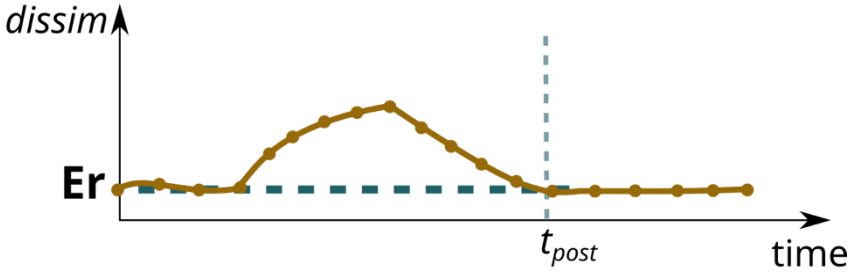
```
recovery_rate(
  type = "compositional",
  metric_tf = c(28, 56),
  ...)
```

Low	High
0.005	0.007

recovery_extent()



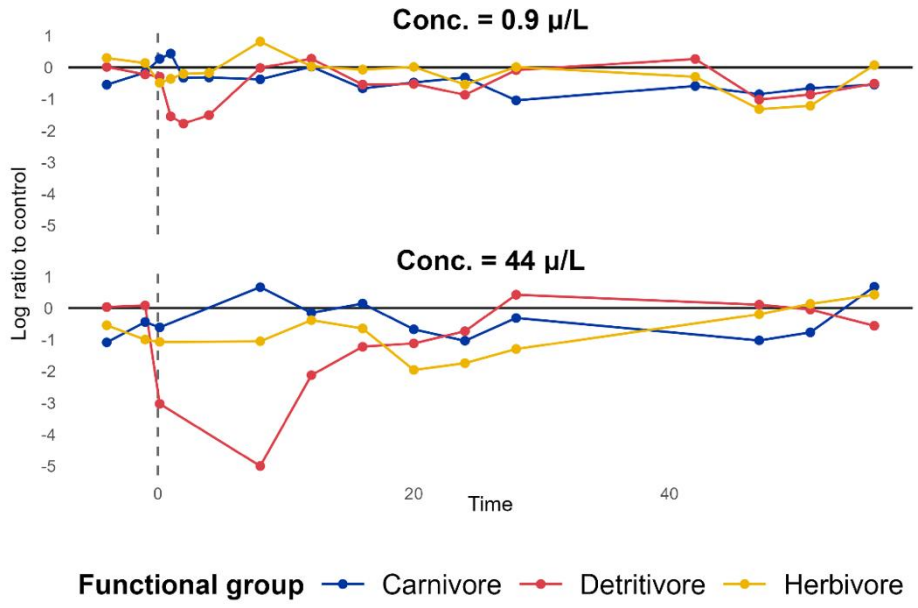

```
recovery_extent(
        type = "functional",
        response = "lrr",
        b = "input",
        t_rec = 28,
        ...)
```




```
recovery_extent(
        type = "compositional",
        t_rec = 28,
        ...)
```

recovery_extent()

Interpretation: Maximum recovery at 0. Smaller or higher values indicate under- or overcompensation



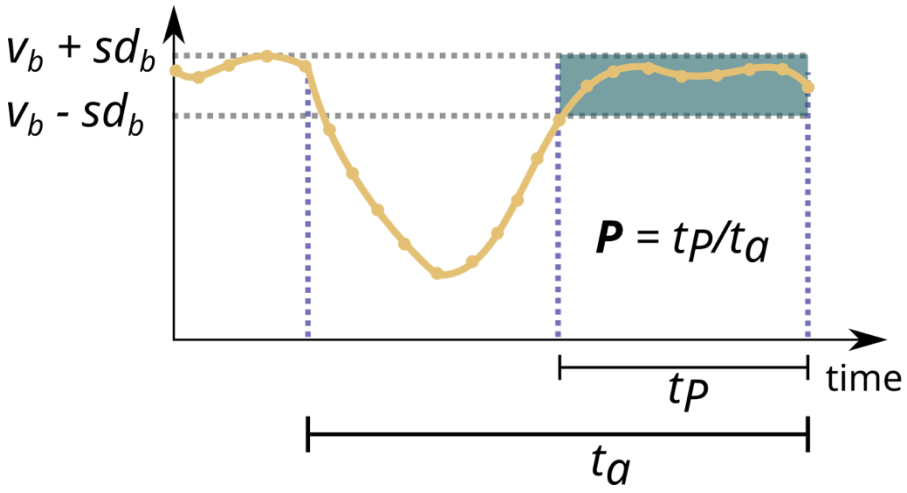

```
recovery_extent(
  type = "functional",
  response = "lrr",
  t_rec = 28,
  ...)
```

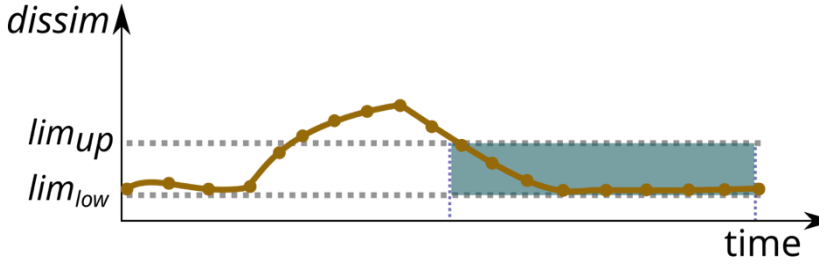
Group	Low	High
Carn	-1.04	-0.31
Detr	-0.08	0.42
Herb	0.01	-1.30

```
recovery_extent(
  type = "compositional",
  t_rec = 28,
  ...)
```

Low	High
0.37	0.24

persistence()



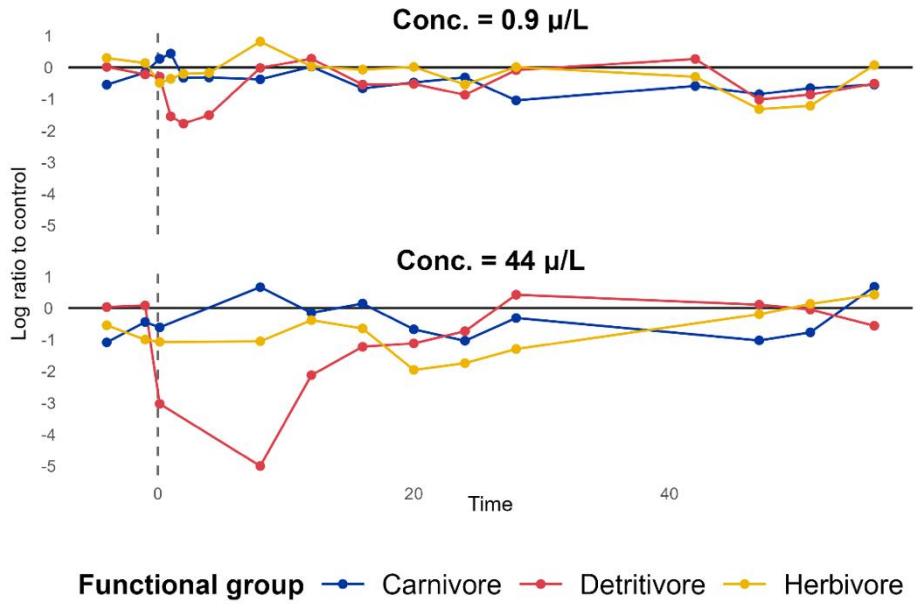



```
persistence(
        type = "functional",
        b = "input",
        metric_tf = c(28, 56),
        ...)
```

```
persistence(
        type = "compositional,
        metric_tf = c(28, 56),
        low_lim = 0.5,
        high_lim = 0.9
        ...)
```

persistence()

Interpretation: *$0 \leq P \leq 1$*. The higher the persistence is, the more stable the system is.



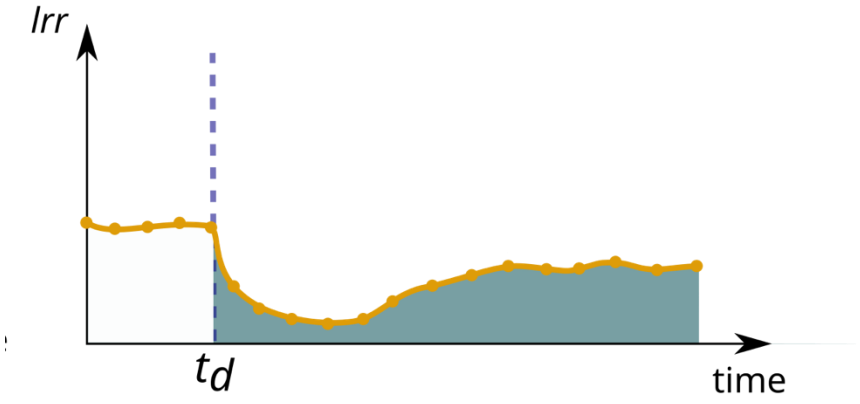

```
persistence(
  type = "functional",
  b = "input",
  metric_tf = c(28, 56),
  ...)
```

Group	Low	High
Carn	0.44	0.6
Detr	0.56	0.52
Herb	0.65	0.9

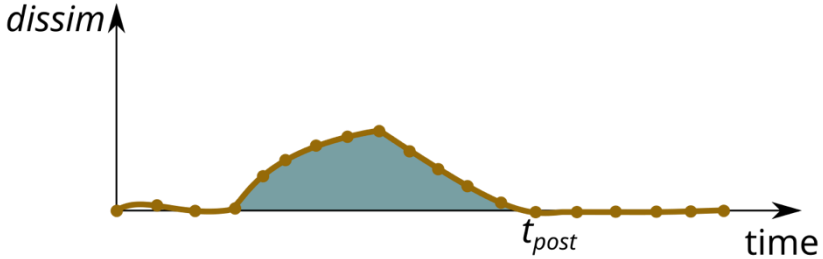
```
persistence(
  type = "compositional,
  metric_tf = c(28, 56),
  low_lim = 0.5,
  high_lim = 0.9,
  ...)
```

Low	High
0.3	0.28

oev()



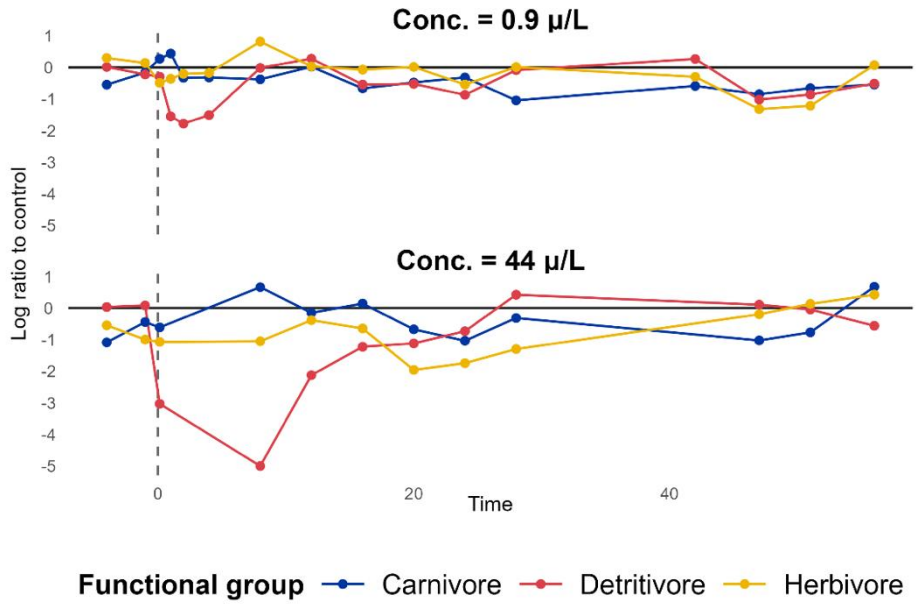

```
oev(
  type = "functional",
  response = "lrr",
  metric_tf = c(0.14, 56),
  …)
```




```
oev(
  type = "compositional",
  metric_tf = c(0.14, 56),
  …)
```

oev()

Interpretation: *OEV ≥ 0, t*he higher the value, the less stable the system.




```
oev(
  type = "functional",
  response = "lrr",
  metric_tf = c(0.14, 56),
  …)
```

Group	Low	High
Carn	31.17	21.85
Detr	23.32	27.27
Herb	21.34	43.84

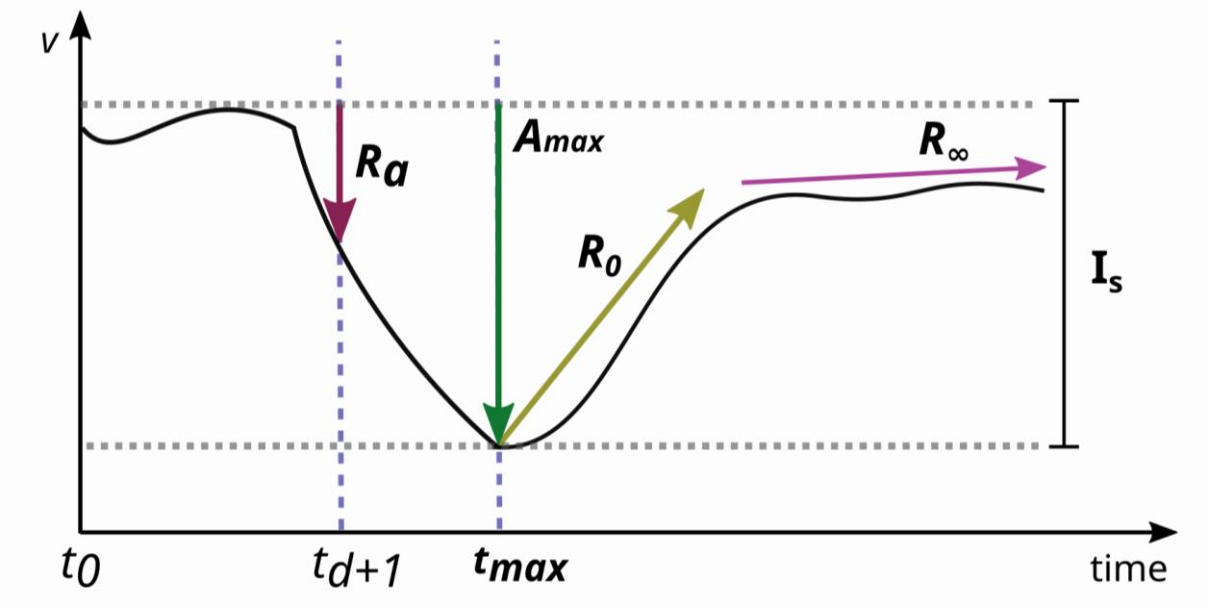
```
oev(
  type = "compositional",
  metric_tf = c(0.14, 56),
  …)
```

Low	High
20.84	24.53

Jacobian metrics

Measure short and long-term dynamics, based on interactions

- Reactivity (R_a)
- Maximal amplification (A_{max})
- Initial resilience (R_0)
- Asymptotic resilience (R_∞)
- Stochastic variability (I_S)



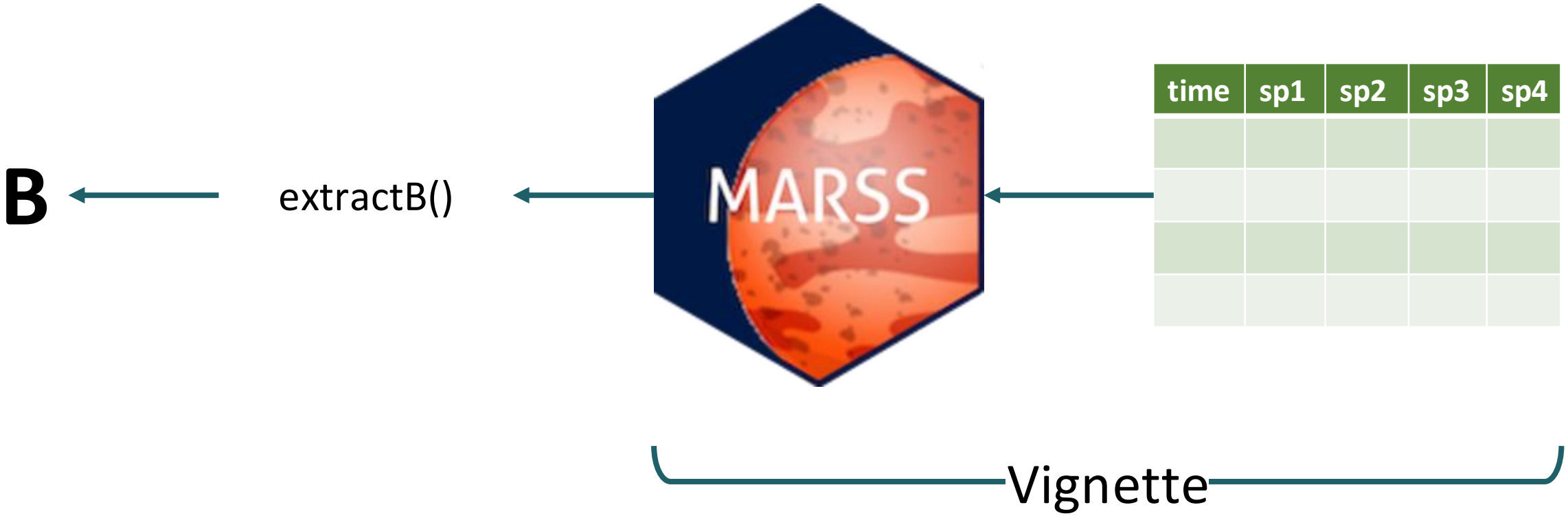
Data input

$$\mathbf{B} = \begin{bmatrix} \boldsymbol{\beta}_{1,1} & \boldsymbol{\beta}_{1,2} & \boldsymbol{\beta}_{1,3} \\ \boldsymbol{\beta}_{2,1} & \boldsymbol{\beta}_{2,2} & \boldsymbol{\beta}_{2,3} \\ \boldsymbol{\beta}_{3,1} & \boldsymbol{\beta}_{3,2} & \boldsymbol{\beta}_{3,3} \end{bmatrix}$$

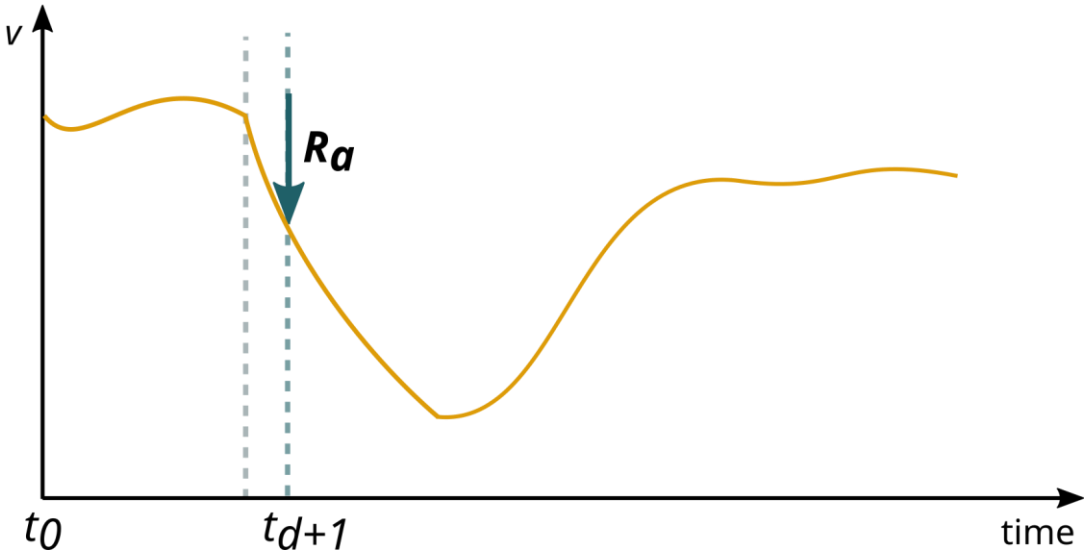
Data input

$$\mathbf{B} = \begin{array}{c|ccc} & \text{sp1} & \text{sp2} & \text{sp3} \\ \hline \text{sp1} & \beta_{1,1} & \beta_{1,2} & \beta_{1,3} \\ \text{sp2} & \beta_{2,1} & \beta_{2,2} & \beta_{2,3} \\ \text{sp3} & \beta_{3,1} & \beta_{3,2} & \beta_{3,3} \end{array}$$

Data input



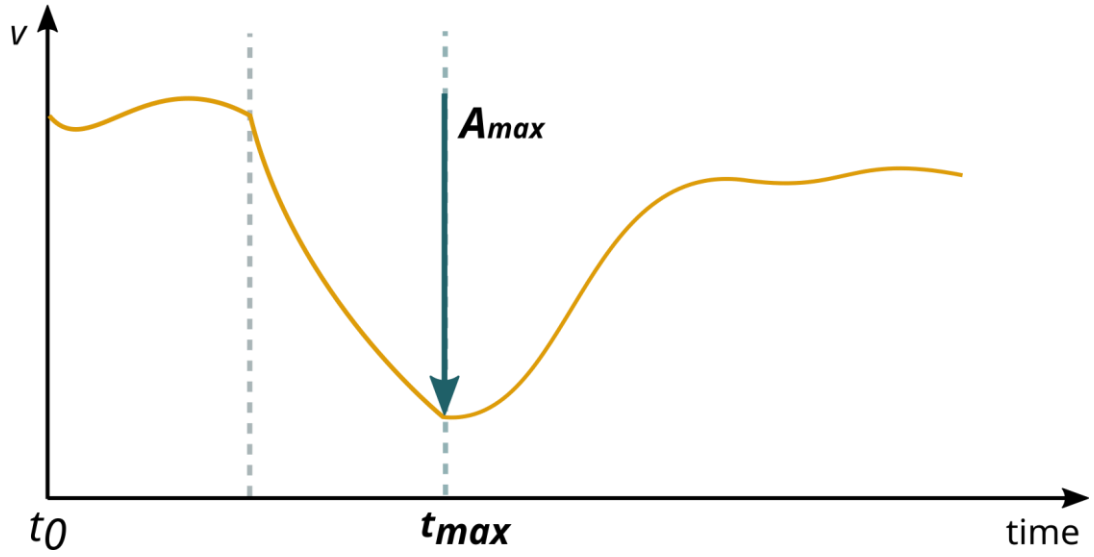
reactvity(B)



Maximum instantaneous rate at which a perturbation can grow at the moment of disturbance.

Interpretation: If $Ra > 0$, the system is initially destabilized. If $Ra < 0$, it isn't.

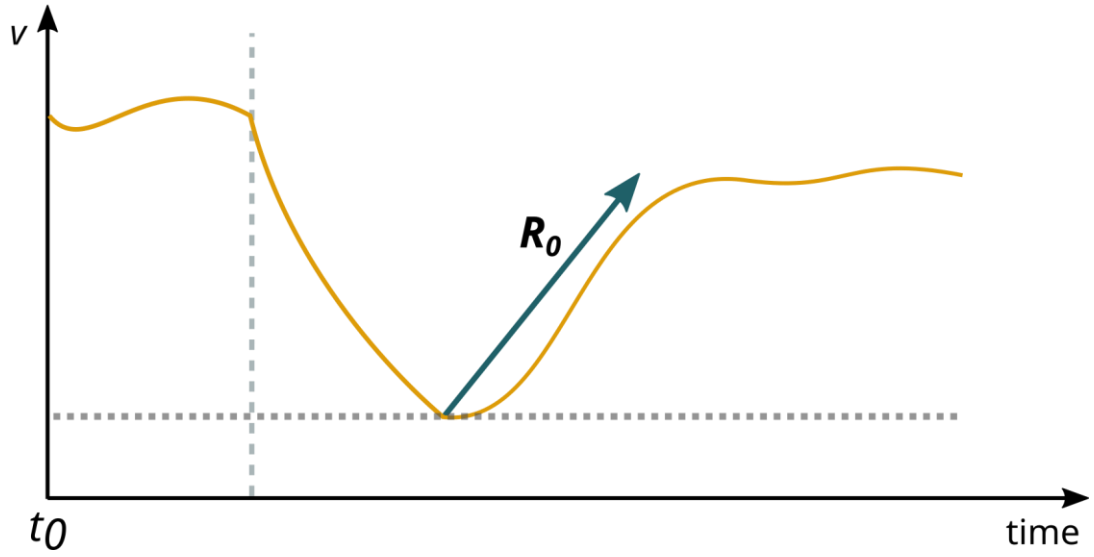
max_amp(B)



Largest factor by which a perturbation is amplified.

Interpretation: If $Amax \approx 1$, the system is stable; if $Amax > 1$, the system overreacts and departs from Equilibrium.

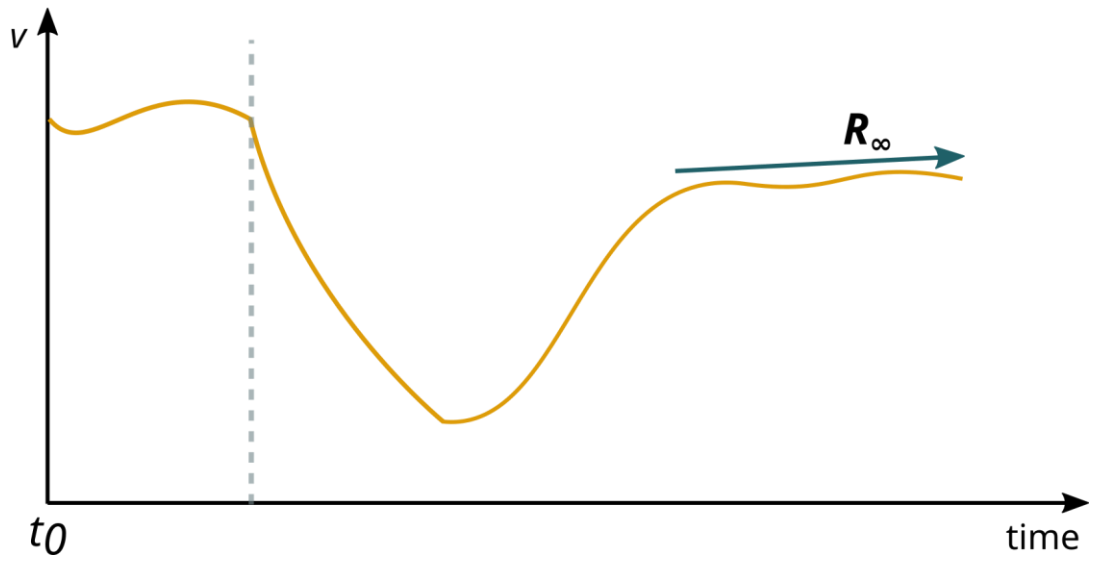
init_resil(B)



Initial rate of return to equilibrium

Interpretation: The larger its value, the more stable the system

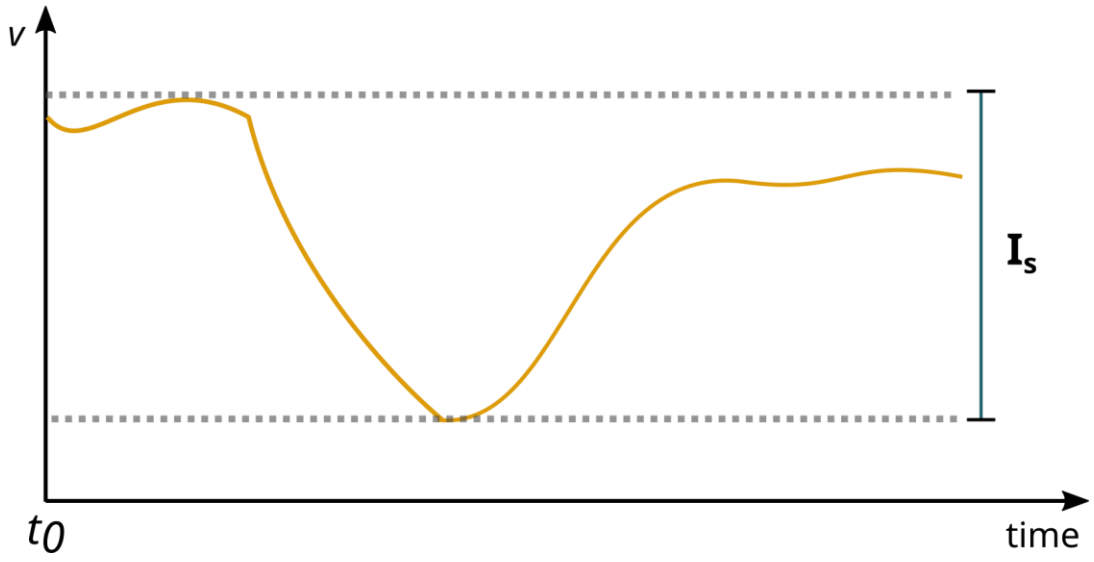
asympt_resil(B)



Slowest/long-term asymptotic rate of return to equilibrium after a pulse perturbation

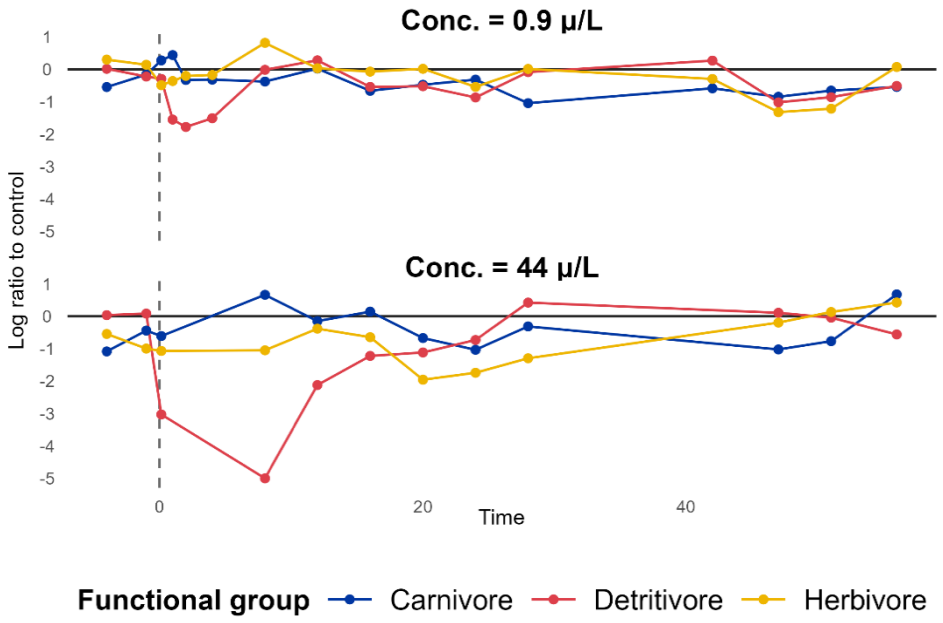
Interpretation: The larger, the more stable the system

stoch_invar(B)



Inverse of the maximal response variance to white noise.

Interpretation: The larger its value, the more stable the system.



	Low	High
Reactivity	0.84	1.03
Maximal amplification	2.99	3.39
Initial resilience	0.39	0.22
Asymptotic resilience	0.196	0.08
Stochastic variability	18.72	19.79

MARSS
and what to beware of

State-space models

- Multivariate autoregressive state-space models

$\mathbf{X}_t = \mathbf{B} \cdot \mathrm{x}_{t-1} + \mathbf{u} + \mathbf{w}_t$, where $\mathbf{w}_t \sim \mathrm{MVN}(0, \mathbf{Q})$

State process (hidden)

$\mathbf{Y}_t = \mathbf{Z} \cdot \mathrm{x}_{t-1} + \mathbf{a} + \mathbf{v}_t$, where $\mathrm{v}_t \sim \mathrm{MVN}(0, \mathbf{R})$

Observation process

State-space models

- Multivariate autoregressive state-space models

$\mathbf{X}_t = \mathbf{B} \cdot \mathrm{x}_{t-1} + \mathbf{u} + \mathbf{w}_t$, where $\mathbf{w}_t \sim \mathrm{MVN}(0, \mathbf{Q})$

State process (hidden)

1:m species

$$\mathbf{X}_t = \begin{bmatrix} x_{s1,t1} & x_{s1,t2} & \cdots & x_{s1,tT} \\ \vdots & & \ddots & \vdots \\ x_{sm,t1} & x_{sm,t2} & \cdots & x_{sm,tT} \end{bmatrix}$$

1:T time points

Unknown abundance values of *m* species or functional groups (rows) over time *T* (columns)

$\mathbf{Y}_t = \mathbf{Z} \cdot \mathrm{x}_{t-1} + \mathbf{a} + \mathbf{v}_t$, where $\mathrm{v}_t \sim \mathrm{MVN}(0, \mathbf{R})$

Observation process

1:n obs

$$\boldsymbol{Y}_t = \begin{bmatrix} y_{1,t1} & y_{1,t2} & \cdots & y_{1,tT} \\ \vdots & & \ddots & \vdots \\ y_{n,t1} & y_{n,t2} & \cdots & y_{n,tT} \end{bmatrix}$$

1:T time points

Matrix of *n* observations of species or functional groups (rows) over time *T* (columns)

$n \geq m$

State process

$$\mathbf{X}_t = \mathbf{B} \cdot \mathrm{x}_{t-1} + \mathbf{u} + \mathbf{w}_t, \text{ where}$$

$$\mathbf{w}_t \sim \mathrm{MVN}(0, \mathbf{Q})$$

1:m species

$$\mathbf{X}_t = \begin{bmatrix} x_{s1,t1} & \cdots & x_{s1,tT} \\ \vdots & \ddots & \vdots \\ x_{sm,t1} & \cdots & x_{sm,tT} \end{bmatrix}$$

1:T time points

A matrix of hidden process abundances of *m* species over time *T*

1:m species

$$\mathbf{B} = \begin{bmatrix} \beta_{1,1} & \cdots & \beta_{1,m} \\ \vdots & \ddots & \vdots \\ \beta_{m,1} & \cdots & \beta_{m,m} \end{bmatrix}$$

Density dependence

1:m species

Interaction matrix **B**

$$\mathbf{u} = \begin{matrix} u_1 \\ u_2 \\ \cdots \\ u_m \end{matrix}$$

Vector of intrinsic rates of increase for *m* species
Often assumed to be 0 for a stationary process

$$\mathbf{w}_t = \begin{matrix} w_1 \\ w_2 \\ \cdots \\ w_m \end{matrix}$$

Vector of process errors that reflects environmental stochasticity

$$\mathbf{w}_t \sim \mathrm{MVN}\left(0, \begin{bmatrix} q_{1,1} & \cdots & q_{1,m} \\ \vdots & \ddots & \vdots \\ q_{m,1} & \cdots & q_{m,m} \end{bmatrix}\right)$$

Observation process

$$\mathbf{Y}_t = \mathbf{Z} \cdot \mathrm{x}_{t-1} + \mathbf{a} + \mathbf{v}_t, \text{ where}$$

$$\mathrm{v}_t \sim \mathrm{MVN}(0, \mathbf{R})$$

1:n obs

$$\boldsymbol{Y_t} = \begin{bmatrix} y_{1,t1} & y_{1,t2} & \cdots & y_{1,tT} \\ \vdots & & \ddots & \vdots \\ y_{n,t1} & y_{n,t2} & \cdots & y_{n,tT} \end{bmatrix}$$

1:T time points

A matrix of *n* observations over time *T*

1:n obs

$$\mathbf{Z} = \begin{bmatrix} 1 & \cdots & 0 \\ \vdots & 1 & \vdots \\ 0 & \cdots & 1 \end{bmatrix}$$

1:m species

Association matrix between *y* and *x*. If 1 observation per hidden process, then *n*=*m* and **Z** is *m* x *m* identity matrix

$$\mathbf{a} = \begin{matrix} a_1 \\ a_2 \\ \cdots \\ a_m \end{matrix}$$

Vector of biases that offset observations from process values

$$\mathbf{v_t} = \begin{matrix} v_1 \\ v_2 \\ \cdots \\ v_m \end{matrix}$$

Vector of observation errors

$$\mathbf{v_t} \sim \mathrm{MVN}(0, \begin{bmatrix} r_{1,1} & \cdots & r_{1,n} \\ \vdots & \ddots & \vdots \\ r_{n,1} & \cdots & r_{n,n} \end{bmatrix})$$

State process: things can be more complex

$$\mathbf{X}_t = \mathbf{B} \cdot x_{t-1} + \mathbf{u} + \mathbf{C} \cdot \mathbf{c}_t + \mathbf{w}_t, \text{ where}$$

1:m species

$$\mathbf{B} = \begin{bmatrix} \beta_{1,1} & \cdots & \beta_{1,m} \\ \vdots & \ddots & \vdots \\ \beta_{m,1} & \cdots & \beta_{m,m} \end{bmatrix}$$

1:m species

Interaction matrix **B**

$$\mathbf{u} = \begin{matrix} u_1 \\ u_2 \\ \cdots \\ u_m \end{matrix}$$

Vector of intrinsic rates of increase for *m* species

1:m species

$$\mathbf{C} = \begin{bmatrix} \gamma_{1,1} & \cdots & \gamma_{1,p} \\ \vdots & \ddots & \vdots \\ \gamma_{m,1} & \cdots & \gamma_{m,p} \end{bmatrix}$$

1:p env cov

Coefficients of the effects *p* covariates have on *m* species

c_1 rain

c_2 food

$\cdots$

c_p temperature

px1 vector of environmental covariate values

$$\mathbf{w}_t \sim \mathrm{MVN}(0, \mathbf{Q})$$

$$\mathbf{w}_t = \begin{matrix} w_1 \\ w_2 \\ \cdots \\ w_m \end{matrix}$$

Vector of process errors

Similarly, we can have covariates included on observation process

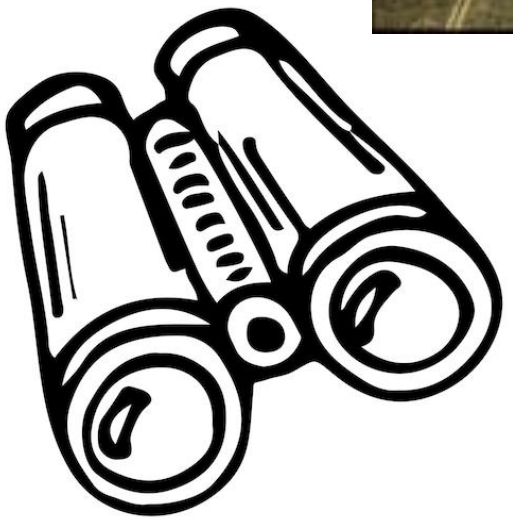
$$\mathbf{w}_t \sim \mathrm{MVN}\left(0, \begin{bmatrix} q_{1,1} & \cdots & q_{1,m} \\ \vdots & \ddots & \vdots \\ q_{m,1} & \cdots & q_{m,m} \end{bmatrix}\right)$$

These are residual process var-covariances after accounting for the effects of included environmental covariates

Multiple observations per process

$$n \geq m$$

- Several replicates per system (e.g. spatial)
- One system observed with different methods (aerial survey vs ground-based one)



Multiple observations per process

$$n \geq m$$

We have a community with 3 functional groups
And 2 replicates per community
m = 3; *n* = 6

1:n obs

$$\mathbf{Z} = \begin{bmatrix} x & \cdots & x \\ \vdots & x & \vdots \\ x & \cdots & x \end{bmatrix}$$

1:m species

1:n obs

$$\mathbf{Z} = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \end{bmatrix}$$

1:m species

Multiple observations per process

$$n \geq m$$

We have a community with 3 functional groups
And 2 replicates per community
m = 3; *n* = 6

1:n obs

$$\mathbf{Z} = \begin{bmatrix} x & \cdots & x \\ \vdots & x & \vdots \\ x & \cdots & x \end{bmatrix}$$

1:m species

1:n obs

$$\mathbf{Z} = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \end{bmatrix}$$

1:m species

R ="diagonal and equal"

1:n obs

$$\mathbf{R} = \begin{bmatrix} r & 0 & 0 & 0 & 0 & 0 \\ 0 & r & 0 & 0 & 0 & 0 \\ 0 & 0 & r & 0 & 0 & 0 \\ 0 & 0 & 0 & r & 0 & 0 \\ 0 & 0 & 0 & 0 & r & 0 \\ 0 & 0 & 0 & 0 & 0 & r \end{bmatrix}$$

1:n obs

R = "diagonal and unequal"

1:n obs

$$\mathbf{R} = \begin{bmatrix} r_1 & 0 & 0 & 0 & 0 & 0 \\ 0 & r_2 & 0 & 0 & 0 & 0 \\ 0 & 0 & r_3 & 0 & 0 & 0 \\ 0 & 0 & 0 & r_4 & 0 & 0 \\ 0 & 0 & 0 & 0 & r_5 & 0 \\ 0 & 0 & 0 & 0 & 0 & r_6 \end{bmatrix}$$

1:n obs

Choices when fitting models

Q – var-covariance matrix of process errors

In MARSS speak

$$\mathbf{Q} = \begin{bmatrix} \sigma_1^2 & \sigma_{1,2} & \sigma_{1,3} \\ \sigma_{1,2} & \sigma_2^2 & \sigma_{2,3} \\ \sigma_{1,3} & \sigma_{2,3} & \sigma_3^2 \end{bmatrix}$$

Q="unconstrained"

$$\mathbf{Q} = \begin{bmatrix} \sigma^2 & 0 & 0 \\ 0 & \sigma^2 & 0 \\ 0 & 0 & \sigma^2 \end{bmatrix}$$

Q="diagonal and equal"

$$\mathbf{Q} = \begin{bmatrix} \sigma_1^2 & 0 & 0 \\ 0 & \sigma_2^2 & 0 \\ 0 & 0 & \sigma_3^2 \end{bmatrix}$$

Q="diagonal and unequal"

$$\mathbf{Q} = \begin{bmatrix} \sigma^2 & q & q \\ q & \sigma^2 & q \\ q & q & \sigma^2 \end{bmatrix}$$

Q="equalvarcov"

Choices when fitting models

Q – var-covariance matrix of process errors

In MARSS speak

$$\mathbf{Q} = \begin{bmatrix} \sigma_1^2 & \sigma_{1,2} & \sigma_{1,3} \\ \sigma_{1,2} & \sigma_2^2 & \sigma_{2,3} \\ \sigma_{1,3} & \sigma_{2,3} & \sigma_3^2 \end{bmatrix}$$

Q="unconstrained"

$$\mathbf{Q} = \begin{bmatrix} \sigma^2 & 0 & 0 \\ 0 & \sigma^2 & 0 \\ 0 & 0 & \sigma^2 \end{bmatrix}$$

Q="diagonal and equal"

$$\mathbf{Q} = \begin{bmatrix} \sigma_1^2 & 0 & 0 \\ 0 & \sigma_2^2 & 0 \\ 0 & 0 & \sigma_3^2 \end{bmatrix}$$

Q="diagonal and unequal"

$$\mathbf{Q} = \begin{bmatrix} \sigma^2 & q & q \\ q & \sigma^2 & q \\ q & q & \sigma^2 \end{bmatrix}$$

Q="equalvarcov"

R – var-covariance matrix of observation errors

$\mathbf{R} = 0$,
full knowledge of the system;
no hidden process

$$\mathbf{R} = \begin{bmatrix} 0.04 & 0 & 0 \\ 0 & 0.04 & 0 \\ 0 & 0 & 0.15 \end{bmatrix}$$

Choices when fitting models

Q – var-covariance matrix of process errors

In MARSS speak

$$\mathbf{Q} = \begin{bmatrix} \sigma_1^2 & \sigma_{1,2} & \sigma_{1,3} \\ \sigma_{1,2} & \sigma_2^2 & \sigma_{2,3} \\ \sigma_{1,3} & \sigma_{2,3} & \sigma_3^2 \end{bmatrix}$$

Q="unconstrained"

$$\mathbf{Q} = \begin{bmatrix} \sigma^2 & 0 & 0 \\ 0 & \sigma^2 & 0 \\ 0 & 0 & \sigma^2 \end{bmatrix}$$

Q="diagonal and equal"

$$\mathbf{Q} = \begin{bmatrix} \sigma_1^2 & 0 & 0 \\ 0 & \sigma_2^2 & 0 \\ 0 & 0 & \sigma_3^2 \end{bmatrix}$$

Q="diagonal and unequal"

$$\mathbf{Q} = \begin{bmatrix} \sigma^2 & q & q \\ q & \sigma^2 & q \\ q & q & \sigma^2 \end{bmatrix}$$

Q="equalvarcov"

R – var-covariance matrix of observation errors

$\mathbf{R} = 0$,
full knowledge of the system;
no hidden process

$$\mathbf{R} = \begin{bmatrix} 0.04 & 0 & 0 \\ 0 & 0.04 & 0 \\ 0 & 0 & 0.15 \end{bmatrix}$$

B – interaction matrix

$$\mathbf{B} = \begin{bmatrix} \beta_{1,1} & \beta_{1,2} & \beta_{1,3} \\ \beta_{2,1} & \beta_{2,2} & \beta_{2,3} \\ \beta_{3,1} & \beta_{3,2} & \beta_{3,3} \end{bmatrix}$$

B="unconstrained"

$$\mathbf{B} = \begin{bmatrix} \beta_{1,1} & \beta_{1,2} & \beta_{1,3} \\ \beta_{2,1} & \beta_{2,2} & \mathbf{0} \\ \beta_{3,1} & \mathbf{0} & \beta_{3,3} \end{bmatrix}$$

Think carefully before specifying the model structure! How does your system function and what main processes have to be reflected in the model?

Beware

- Even though it is possible to specify the model with many parameters, such model may not be identifiable
- Convergence issues (e.g. when the model is degenerate, i.e. one of elements on the diagonal of **Q** or **R** are going to 0)
- Time series have to be long enough to estimate parameters
- There should be no missing data in $\mathbf{c}_t$ (environmental covariates' values)
- The user must carefully specify the model (often several alternative ones) that reflect the system in consideration
- Best to simulate a similar dataset and fit MARSS to it to learn what behavior/parameters can be recovered

Thank you for standing with Ukraine